Average, Total, and Concentration Hierarchical Indices

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12 April 2026

Abstract

Hierarchy is pervasive, yet existing measures obscure whether it represents a per-capita condition, an aggregate volume, or a concentration of authority. This paper develops an axiomatic framework for measuring hierarchy on directed acyclic graphs. We prove that no regular index can combine separability with unrestricted extension axioms. Resolving this tension, we characterize three mutually exclusive families. Average bilateral indices quantify per-capita hierarchy and remain invariant under replication. Total bilateral indices measure aggregate structural volume and scale linearly. Concentration indices capture authority density and decline under replication. The framework clarifies the normative choice between intensive, extensive, and concentration-based paradigms for measurement.

Keywords: hierarchy measurement, hierarchical index, directed acyclic graph, additive separability, organizational structure.
JEL classifications: D23, L22.

1 Introduction

Hierarchy shapes behavior across animal and human societies, regulating resource access, power dynamics, and social coordination. Whether formalized in organizational charts, embedded in informal networks, or constructed through status norms, hierarchical structures

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permeate nearly every domain of human interaction. High-status actors often exhibit greater confidence and influence, while those lower in the order face constraints that shape conformity, competition, and mobility.

These structures generate complex outcomes. Hierarchies can enhance stability, coordination, and efficiency in uncertain environments (Simon, 1962, 1973; Holland, 1998; Lawrence and Lorsch, 1967; Thompson, 1967; Lincoln and Kalleberg, 1985), and informal networks often bolster organizational resilience, profitability, and crisis management (Krackhardt and Stern, 1988; Sarkar et al., 2010; Culen, 2017). Yet the same architectures can enable power abuse, entrench inequality, and stifle innovation (Anderson, 1972; Fix, 2017, 2018, 2019, 2021; Wright, 2024). Hierarchy is thus not a passive backdrop but an active mechanism that channels social and economic outcomes.

Precise measurement is therefore essential. While prior axiomatic work has established ordinal frameworks via incomplete pre-orders, existing cardinal indices lack formal structural grounding. This leaves a clear gap for measures that yield complete rankings and are characterized by structural axioms. To address this, we develop a new class of hierarchical indices grounded in a significantly expanded axiomatic framework. Building on the ordinal foundations of Carbonell-Nicolau (2025a,b)—which relied on anonymity, replication invariance, and subordination removal—we introduce novel replication principles and vertical extension axioms. Whereas the prior framework leaves many structurally distinct hierarchies incomparable, particularly under depth transformations, our expanded axiomatization resolves this limitation, yielding complete cardinal rankings that capture structural dimensions systematically overlooked by conventional measures.

A central conceptual contribution is the recognition that measuring hierarchy requires a fundamental normative choice about how the concept should scale with organizational size. This choice manifests in three mutually incompatible visions: hierarchy as an extensive cumulative volume, an intensive per-capita average, or a concentrated global density. Each vision is anchored by a distinct replication principle dictating whether cloning an organization strictly increases, preserves, or dilutes the measured hierarchy. To ensure mathematical consistency, we formalize three corresponding separability structures that differ exclusively in population normalization: absolute additivity (no scaling), linear averaging (division by population size), and quadratic density scaling (division by the square of population size). Selecting a conceptual paradigm simultaneously fixes the index's replication behavior and the separability structure required to sustain it. No single index can satisfy more than one pathway.

Beyond replication, a complete measurement framework must account for vertical growth and structural transformation. We therefore introduce axioms formalizing how appending management layers—above or below an existing structure—affects the overall measurement. Strict extension principles require that any valid vertical attachment increases the hierarchy score, while relaxed variants accommodate partial linkages provided the appended structure is itself sufficiently hierarchical.

Our analysis begins with two impossibility results. We prove that no regular hierarchical index can simultaneously satisfy per-capita or concentration-based scaling alongside unrestricted vertical extension. To resolve this tension, we characterize three complementary families of indices. The first identifies intensive, per-capita measures that remain invariant under replication but satisfy only relaxed extension conditions. The second identifies extensive, additive measures that scale linearly with population and satisfy unrestricted extension. The third identifies concentration measures that penalize structural fragmentation and strictly decrease under replication. Each family is uniquely pinned down by its normalization rule and a tailored population-growth condition ensuring meaningful asymptotic behavior.

Applying this framework to established indices reveals systematic axiom violations, underscoring the need for rigorous structural validation. The global reaching centrality index (Mones et al., 2012), widely used in network science, satisfies anonymity but its reliance on a global reference point destroys separability and generates non-monotonic behavior, violating core extension and subordination removal axioms. Krackhardt's (1994) four dimensions of informal organization prove either trivial or axiomatically incompatible in our framework. Other measures face similar limitations: indices by Trusina et al. (2004), Corominas-Murtra et al. (2013), and Luo and Magee (2011) are calibrated to detect cycles, collapsing to constant values on acyclic graphs. Finally, the algorithmic approach of Czégel and Palla (2015) depends on an exogenous decay parameter; this sensitivity can violate extension axioms unless optimally tuned, limiting its robustness for structural comparison.

The remainder of the paper proceeds as follows. Section 2 formalizes hierarchies as directed acyclic graphs and introduces the structural primitives used throughout the analysis. Section 3 develops the expanded axiomatic framework, detailing the replication principles and vertical extension axioms that ground our measurement approach. Section 4 presents the core theoretical results, establishing the impossibility of combining normalized separability with unrestricted extension and providing complete characterizations for the average, total, and concentration index families. Section 5 applies the framework to prominent indices from the literature, demonstrating their systematic failure to satisfy core structural axioms. Section 6

concludes by discussing the implications of the normative trichotomy for organizational measurement and outlining directions for empirical application and theoretical extension.

2 Hierarchies

A *hierarchy* is a *directed acyclic graph* (DAG) $h = (V(h), E(h))$, where $V(h)$ denotes the set of individuals and $E(h)$ denotes the set of directed edges. An edge $(j, i) \in E(h)$ indicates that j is a (direct) supervisor of i. Acyclicity ensures that following edges in their direction never returns to the starting node; equivalently, there are no directed cycles.

For $v \in V(h)$, let

$$\mathrm{Anc}_h(v) = \{u \in V(h) : u \rightsquigarrow v \text{ via a directed path of length } \geq 1\}$$

and

$$\mathrm{Desc}_h(v) = \{w \in V(h) : v \rightsquigarrow w \text{ via a directed path of length } \geq 1\}$$

denote the sets of (direct or indirect) supervisors and subordinates of v, respectively. We define the *depth* of v as

$$s_h(v) = |\mathrm{Anc}_h(v)|,$$

i.e., the total number of (direct or indirect) supervisors of v. Symmetrically, we define the *span* of v as

$$\sigma_h(v) = |\mathrm{Desc}_h(v)|,$$

the total number of (direct or indirect) subordinates of v.

This structure induces a unidirectional (downward) flow of authority. Individuals may have direct supervisors (immediate predecessors) and indirect supervisors (supervisors of their supervisors), producing transitive chains of command. The DAG assumption guarantees that the set of reporting lines is well-defined (acyclic), though not necessarily unique when multiple supervisors exist.

Because the graph is acyclic, an individual's depth is strictly greater than the depth of any of their supervisors: for any directed edge $j \to i$ in h, we have

$$\mathrm{Anc}_h(j) \subseteq \mathrm{Anc}_h(i) \quad \text{and} \quad j \in \mathrm{Anc}_h(i) \setminus \mathrm{Anc}_h(j),$$

hence $s_h(j) < s_h(i)$. Thus, if an individual i has k total supervisors, each of those supervisors

has strictly fewer than k supervisors.

A node with no supervisors (in-degree 0) is a *root node*; a node with no subordinates (out-degree 0) is a *leaf node*. The *size* of a hierarchy is the total number of nodes, $|V(h)|$.

Let $\mathscr{H}$ denote the class of all (finite) hierarchies. For $n \geq 1$, write $\mathscr{H}_n = \{ h \in \mathscr{H} : |V(h)| = n \}$, so that

$$\mathscr{H} = \bigcup_{n \geq 1} \mathscr{H}_n.$$

A *composite hierarchy* $(h_1, \ldots, h_k)$, with $k \geq 1$, is an ordered k-tuple of pairwise node-disjoint hierarchies; we identify it with their disjoint union $h_1 \cup \cdots \cup h_k$.

3 Axiomatic framework

As established in Section 2, let $\mathscr{H}$ denote the universal class of finite hierarchies (directed acyclic graphs). The approach in Carbonell-Nicolau (2025a,b) formalizes hierarchy via an *ordinal* pre-order on $\mathscr{H}$. In this paper, our primary object is a *cardinal hierarchical index*, a real-valued mapping $I : \mathscr{H} \to \mathbb{R}$ that induces a complete pre-order by numerical ranking: $h' \succcurlyeq h \Leftrightarrow I(h') \geq I(h)$. The symmetric part ($\sim$) corresponds to equality ($I(h') = I(h)$), and the asymmetric part ($\succ$) corresponds to strict inequality ($I(h') > I(h)$).

A hierarchy h' is a *relabeling* (isomorphic copy) of h if there exists a bijection $\phi : V(h) \to V(h')$ such that, for all $j, i \in V(h)$,

$$(j, i) \in E(h) \Leftrightarrow (\phi(j), \phi(i)) \in E(h').$$

Thus, h and h' share the same authority structure and differ only in node labels.

Anonymity (A). A hierarchical index $I : \mathscr{H} \to \mathbb{R}$ satisfies A if $I(h') = I(h)$ whenever h' is a relabeling of h.

A *replication* of $h \in \mathscr{H}$ is the disjoint union of $k \geq 1$ identical copies of h. If hierarchy is conceived as an average, per-capita property of an organization, scaling up the population by simply cloning its structure should leave the overall measurement unchanged.

Replication Principle (RP). A hierarchical index $I : \mathscr{H} \to \mathbb{R}$ satisfies RP if $I(h') = I(h)$ whenever h' is a replication of h.

Subordination removal. Let $P_h(j) = \{ p \in V(h) : (p, j) \in E(h) \}$ denote the set of *immediate* supervisors of j.

Definition 1. Let $h \in \mathscr{H}$. We say h' is obtained from h by *removing* a subordination relation $(j, i) \in E(h)$ if:

1. Delete (j, i).
2. If $P_h(j) = \varnothing$ (i.e., j is a root), add no further edges.
3. If $P_h(j) \neq \varnothing$, then for each $p \in P_h(j)$:
   - add the edge (p, i) *unless* there already exists in h a directed path from p to i that does not traverse j;
   - otherwise, make no change.
4. All other supervisory relations in h remain unchanged.

The resulting h' is again a DAG: each new edge (p, i) points from a superior of j to a subordinate of j, so no directed cycle can be created.

An edge $(j, i) \in E(h)$ is *redundant* if there exists a directed path from j to i distinct from (j, i) itself.

Subordination Removal (SR). A hierarchical index $I : \mathscr{H} \to \mathbb{R}$ satisfies SR if $I(h) \geq I(h')$ whenever h' is obtained from h by removing a subordination relation. Moreover, $I(h) > I(h')$ whenever the removed edge is non-redundant in h.

Remark 1 (From pre-orders to indices)**.** The axioms A, RP, and SR originate in the ordinal literature, where they characterize the topological pre-order obtained by successive subordination removals (Carbonell-Nicolau, 2025b). In our cardinal framework, these axioms are imposed directly on I, and subsequent sections develop three complementary families of indices: *average* indices, which evaluate per-capita hierarchical degree and remain invariant under replication; *total* indices, which measure aggregate structural volume and scale linearly with population size; and *concentration* indices, which capture the density of realized authority relations through quadratic normalization. Each family corresponds to a distinct normative paradigm for how hierarchy should scale with organizational size.

3.1 Replication principles and structural scaling

We examine how a hierarchical index I behaves when a hierarchy h is *replicated* into a disjoint composite of $k \geq 1$ identical copies. Formally, h^k denotes the disjoint union of k pairwise node-disjoint hierarchies, each isomorphic to h.

Intuition yields conflicting, antagonistic answers regarding whether for $k > 1$, the composite h^k is equally, more, or less hierarchical than the isolated h. This tension reflects that "hierarchy" conflates at least three distinct structural notions: per-capita shape, aggregate mass, and global density. To disentangle these, we formalize three pairwise incompatible variants of the replication principle.

3.1.1 Hierarchy as an average experience (the intensive view)

The baseline Replication Principle (RP) introduced earlier belongs to the intensive measurement paradigm. If we conceptualize hierarchy as the expected, day-to-day structural experience of a randomly selected node, cloning the society does not alter this average local environment. We re-designate this standard condition as the Intensive Replication Principle.

Intensive Replication Principle (IRP). A hierarchical index $I : \mathscr{H} \to \mathbb{R}$ satisfies IRP if for any hierarchy h and any integer $k \geq 1$, $I(h^k) = I(h)$.

The intuitive appeal of IRP lies in its preservation of intensive structural properties. Replicating a hierarchy scales both the aggregate structural evaluation and the population size proportionally, leaving the per-capita average unchanged. This invariance formalizes the intensive perspective, treating hierarchy as a scale-independent structural condition rather than a cumulative volume.

3.1.2 Hierarchy as an aggregate volume (the extensive view)

In contrast to the intensive perspective, one might view hierarchy not as a per-capita shape, but as a total volume of administrative overhead, systemic friction, or total relational control. From this macroeconomic perspective, 1,000 isolated small businesses contain vastly more aggregate "managerial substance" than a single small business.

Extensive Replication Principle (ERP). A hierarchical index $I : \mathscr{H} \to \mathbb{R}$ satisfies ERP if for any hierarchy h with at least one subordination relation (i.e., $E(h) \neq \varnothing$) and any integer $k > 1$, $I(h^k) > I(h)$.

The ERP captures the intuition that hierarchical relations are additive units of structure. While IRP treats hierarchy as a scale-invariant per-capita property, ERP conceptualizes it as a cumulative structural volume. Under this extensive view, replicating a hierarchy strictly increases the index.

3.1.3 Hierarchy as global centralization (the concentration view)

Finally, intuition frequently suggests a third, antagonistic perspective: that 1,000 disconnected copies of a two-person team is severely *less* hierarchical than a single two-person team. This intuition views hierarchy as a measure of structural concentration or network density.

Concentration Replication Principle (CRP). A hierarchical index $I : \mathscr{H} \to \mathbb{R}$ satisfies CRP if for any hierarchy h with at least one subordination relation (i.e., $E(h) \neq \varnothing$) and any integer $k > 1$, $I(h^k) < I(h)$.

While CRP and ERP track opposite inequalities, both are logically grounded in their implicit normalization baselines. In a system of N nodes, the maximum possible number of supervisory relationships grows quadratically with population size. When a hierarchy is replicated k times ($k > 1$) without introducing cross-links, the actual number of reporting links increases linearly (by a factor of k), while the theoretical capacity for relationships expands quadratically (by a factor of k^2). This structural mismatch inevitably dilutes network density. CRP formalizes the intuition that a large population fragmented into isolated chains of command is inherently less concentrated relative to its maximum possible connectivity.

No single index can satisfy more than one of IRP, ERP, and CRP. Thus, choosing a replication principle explicitly fixes whether an index measures structural averages, structural accumulation, or structural density.

Remark 2 (analogues in socioeconomic measurement)**.** Because network topologies bind the shape of localized relationships to the scale and density of the global population, measuring hierarchy forces a choice between three distinct paradigms that mirror debates across economics:

- **The Intensive View (IRP):** In inequality measurement, this aligns with Dalton's Population Principle (Dalton, 1920), which mandates that replicating a society leaves its inequality invariant. In welfare economics, it parallels Average Utilitarianism, evaluating systems strictly in per-capita terms.
- **The Extensive View (ERP):** In poverty analysis, the aggregate poverty gap captures the total absolute volume of deprivation, scaling linearly with replication. In population ethics, this maps to Classical (Total) Utilitarianism, where adding identical individuals accumulates aggregate societal welfare (Blackorby et al., 2005).
- **The Concentration View (CRP):** The intuition that replication dilutes structure parallels market concentration in Industrial Organization. For instance, the Herfindahl-Hirschman Index (Hirschman, 1964) drops drastically when a monopoly is replicated

into a k-firm oligopoly. Similarly, in theories of fairness (e.g., Fleurbaey and Maniquet, 2011), social evaluation is highly sensitive to the relative concentration of disadvantage; cloning a society dilutes this concentration, thereby altering the macro-level assessment.

Analogous to the concentration view, CRP dictates that replicating a hierarchy fragments a singular concentration of power into competing, isolated structures. Ultimately, by selecting a replication principle, a structural index explicitly commits to one of these three classical perspectives: per-capita shape, absolute volume, or relative density.

3.2 Upward Extension

Conventions. Throughout this and the following two subsections (Upward Extension, Downward Extension, and Additional extension axioms), terms such as 'leaf', 'root', and 'maximum depth' refer to the corresponding notions in the *pre-extension* hierarchies, i.e., before any cross-edges are added. Furthermore, when taking the disjoint union of two hierarchies to form a compound, we preserve all original nodes and edges with no nodes identified or merged.[1] In each compound defined below, the first argument acts as the upper component and the second as the lower component. Newly added cross-edges are exclusively directed from nodes in the upper component to nodes in the lower component. Because the components are disjoint and no cross-edges point back, directed cycles cannot be created, inherently preserving acyclicity.

We now introduce an axiom concerning hierarchical extensions that place one hierarchy *above* another by linking the leaf nodes of the upper hierarchy to the root nodes of the lower hierarchy.

Appended compound (set-valued operator). For $h_o, h \in \mathscr{H}$, let $h_o \oplus h$ denote the *set* of hierarchies obtained by:

1. taking the disjoint union of h_o and h, and
2. adding edges from each leaf node of h_o to one or more root nodes of h so that every leaf in h_o acquires at least one outgoing edge to a root in h.

Only such leaf-to-root edges are added; no other cross-edges are permitted.

[1] If the node sets are not disjoint, we replace the hierarchies with isomorphic copies that have disjoint node labels before taking the union. This standard convention makes the disjoint union well-defined; when I satisfies Anonymity (A), the choice of relabeling is immaterial for evaluation.

Remark 3. Since every finite DAG has at least one leaf and at least one root (by finiteness, an unextendable directed path exists; its start is a root and its end a leaf), $h_o \oplus h \neq \emptyset$ for all $h_o, h \in \mathscr{H}$ (e.g., one can always pick an arbitrary root r of h and add an edge from every leaf of h_o to r). The operator $\oplus$ is *set-valued* and *order-dependent*: in general $h_o \oplus h \neq h \oplus h_o$, and there can be multiple valid compounds in $h_o \oplus h$ depending on how leaves are attached to roots. Note that while every leaf of h_o must connect to h (ensuring the upper structure is fully connected by cross-edges), we intentionally do not require every root of h to receive a connection. The operator formalizes placing a structure above h, which does not strictly require integrating every independent component of the base hierarchy.

Upward Extension (UE). A hierarchical index $I : \mathscr{H} \to \mathbb{R}$ satisfies UE if for all $h_o, h \in \mathscr{H}$ and every $h^* \in h_o \oplus h$, we have $I(h^*) > I(h)$.

Intuitively, appending h beneath h_o strictly increases hierarchical depth, so any resulting compound hierarchy must be evaluated as strictly more hierarchical than h alone. UE intentionally imposes unrestricted monotonicity: even placing a flat hierarchy above h must strictly increase the index, because the axiom formalizes "depth creation" rather than the "quality" of the appended module. This is illustrated in Figure 1, where h' results from appending h beneath a single-node h_o; by UE, $I(h') > I(h)$.

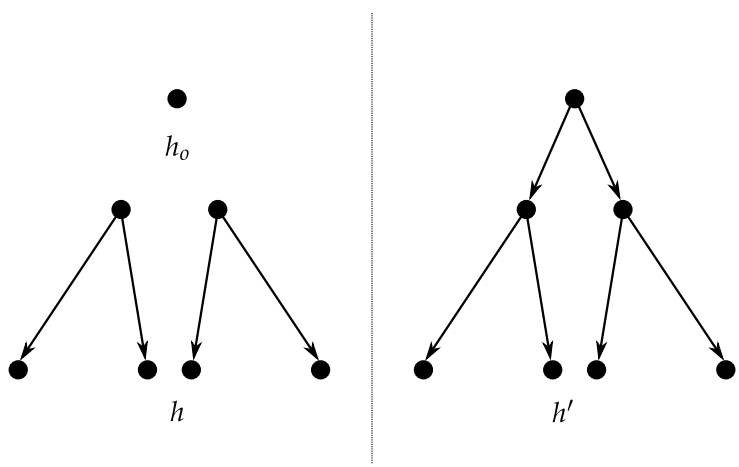

Figure 1: Illustrating UE.

A critical requirement of this operator is that *every* leaf node in h_o must connect to at least one root node in h. This constraint rules out incomplete attachments like the one shown in Figure 2, where h is positioned below a predominantly flat hierarchy h_o, but the

connections are incomplete. Such configurations create structural ambiguity, as the resulting compound hierarchy leaves most of h_o essentially flat and only "anecdotally" attached to h.

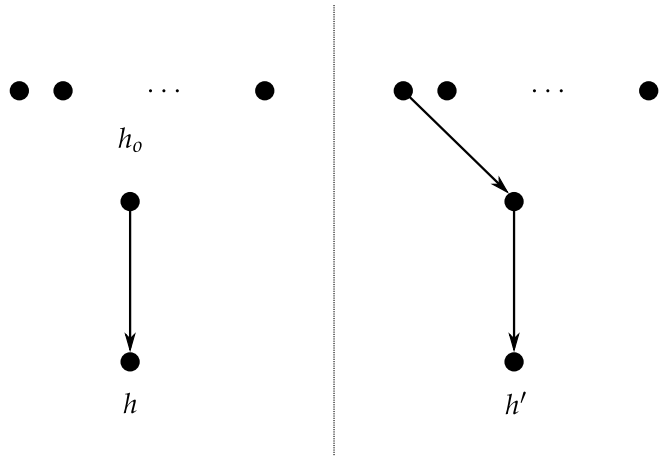

Figure 2: Incomplete attachments are excluded by the definition of $h_o \oplus h$.

The proper construction is demonstrated in Figure 3, where all leaf nodes of h_o attach to a root node of h (in the example, the unique root), yielding $h'' \in h_o \oplus h$. According to UE, $I(h'') > I(h)$—an intuitive conclusion given the increased depth of h''.

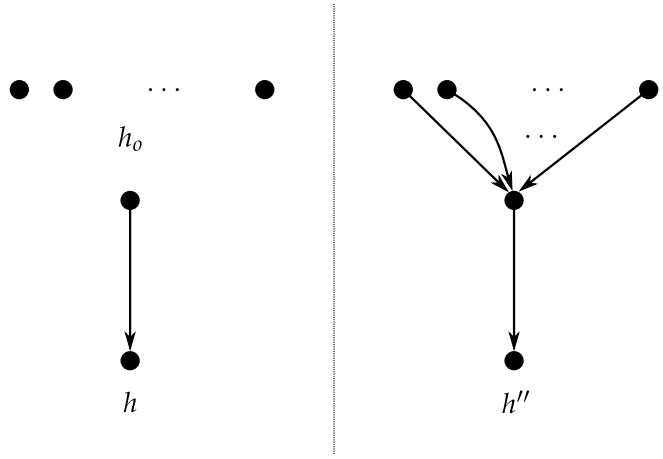

Figure 3: Valid construction in $h_o \oplus h$.

UE versus SR. Upward Extension and Subordination Removal capture distinct, complementary aspects of hierarchical structure. UE responds to vertical *superposition* via the $\oplus$ operator, which enhances a structure's vertical dimension, whereas SR focuses on short-circuiting a single reporting edge.

This complementarity is illustrated in Figure 4, which shows $h' \in h_o \oplus h$ with $I(h') > I(h)$ by UE, yet h' cannot be obtained from h by any sequence of subordination removals. This holds true even after making the node counts comparable by replication. Consider Figure 5, which shows a 3-times replica of h in the left panel and a 2-times replica of h' in the right panel. It remains impossible to obtain the 3-times replica from the 2-times replica via subordination removal.

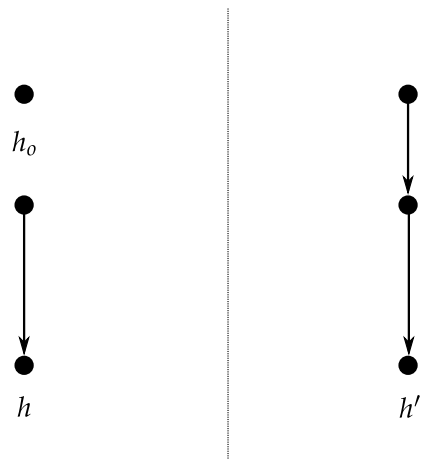

Figure 4: UE versus SR.

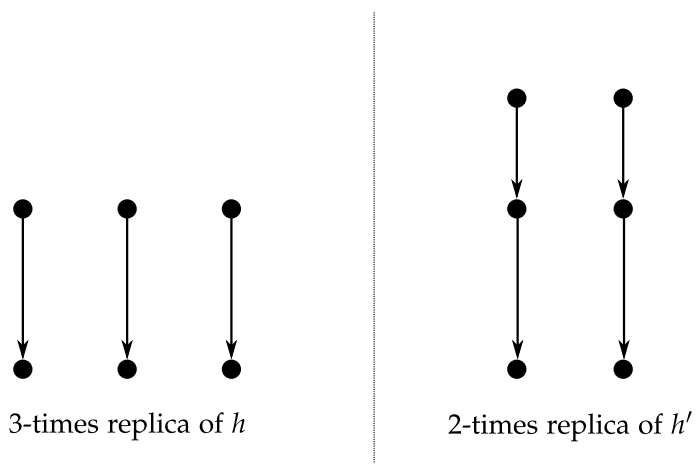

Figure 5: Replicas of h (left) and h' (right) remain unrelated by SR.

Conversely, there are cases where h is obtained from h' by removing a subordination relation (so SR applies), yet h' cannot be derived from h through the appending operator. For a simple illustration, consider the hierarchies depicted in Figure 6, where removing

an edge in h' yields h. However, h' cannot be written as an element of $h_o \oplus h$ for any h_o, because any such $\oplus$-compound would necessarily contain at least four nodes, which this specific h' violates.

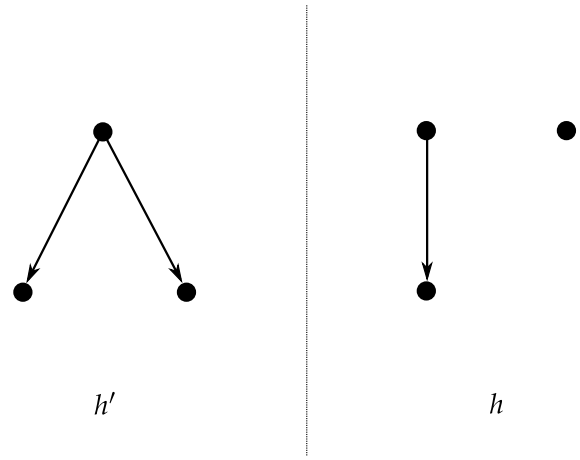

Figure 6: An instance where SR applies but UE does not.

3.3 Downward Extension

We now introduce an axiom for hierarchical extensions that place one hierarchy *below* another by linking (via new edges) the root nodes of the lower hierarchy to leaf nodes of the upper hierarchy that have maximum depth.

Appended compound (set-valued operator). For $h, h_o \in \mathscr{H}$, let $h \boxplus h_o$ denote the *set* of hierarchies obtained by:

1. taking the disjoint union of h and h_o, and
2. for *each* root node r of h_o, adding one or more edges from leaf nodes of h of *maximum depth* to r.

Only such maximum-depth-leaf-to-root edges are added; no other cross-edges are permitted.

Remark 4. Since h is a finite DAG it has at least one leaf of maximal depth (the node of maximum global depth must be a leaf, since if $v \to w$ then $s_h(w) > s_h(v)$), and since h_o is a finite DAG it has at least one root; hence $h \boxplus h_o \neq \emptyset$ for all $h, h_o \in \mathscr{H}$ (e.g., one can always pick a single maximum-depth leaf $\ell^{\max}$ of h and add an edge from $\ell^{\max}$ to every root of h_o). The operator $\boxplus$ is *set-valued* and *order-dependent*: in general $h \boxplus h_o \neq h_o \boxplus h$. Note that maximum depth is evaluated globally across all nodes in h, rather than componentwise.

Furthermore, in contrast to the top-coverage requirement of $\oplus$, while every root of h_o must be connected to h, we do not require every maximum-depth leaf of h to send a connection.

Downward Extension (DE). A hierarchical index $I : \mathscr{H} \to \mathbb{R}$ satisfies DE if for all $h, h_o \in \mathscr{H}$ and every $h^* \in h \boxplus h_o$, we have $I(h^*) > I(h)$.

Intuitively, appending h_o *below* h strictly deepens the hierarchy at its bottom boundary; any compound $h^* \in h \boxplus h_o$ must therefore be evaluated as strictly more hierarchical than h alone. Much like UE, DE captures depth creation rather than module quality, and thus demands a strict increase even if the lower appendage h_o is entirely flat.

We illustrate the operator $\boxplus$ in Figures 7 and 8. In Figure 7, a purely horizontal hierarchy h_o is attached to a leaf of h that is *not* of maximum depth; the resulting compound h' violates the maximum-depth requirement, hence $h' \notin h \boxplus h_o$. By contrast, Figure 8 shows the proper construction: every added edge originates at a maximum-depth leaf of h and points to a root of h_o, so the resulting compound h' is a valid member of $h \boxplus h_o$. By DE, $I(h') > I(h)$.

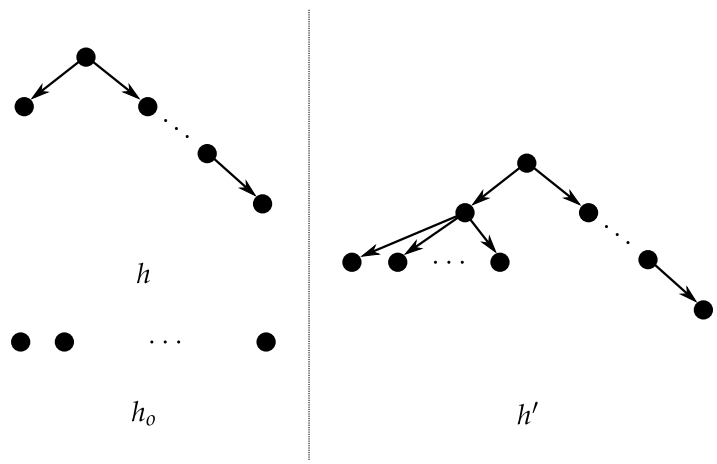

Figure 7: $h' \notin h \boxplus h_o$ (attachment is not at maximum depth).

DE versus UE and SR. There exist pairs of hierarchies that are comparable under DE but incomparable under UE and SR. As shown in Figure 9, we have $h' \in h \boxplus h_o$, yet there is no hierarchy $h^\dagger$ with $h' \in h^\dagger \oplus h$. Moreover, even after replication (to equalize node counts), no sequence of subordination removals transforms the replica of h' into the replica of h. Thus, DE captures vertical growth at the bottom boundary that UE and SR do not.

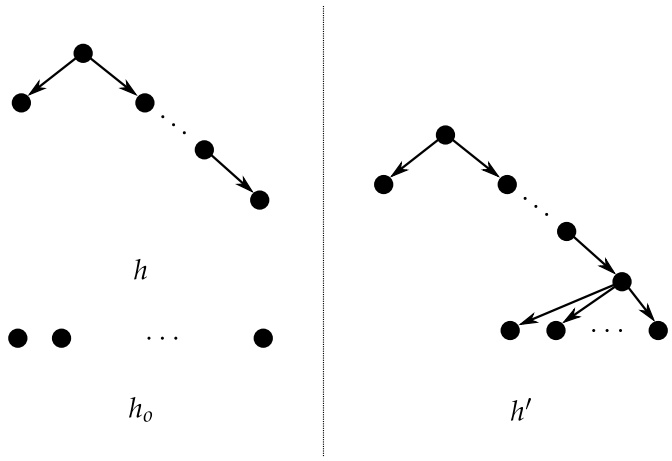

Figure 8: $h' \in h \boxplus h_o$ (all attachments from maximum-depth leaves to roots).

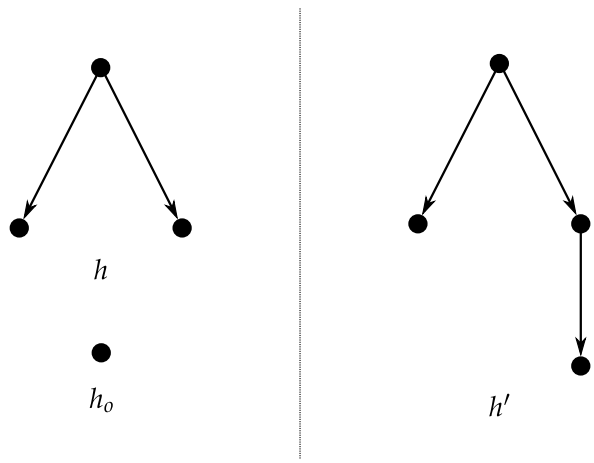

Figure 9: DE versus UE and SR.

3.4 Additional extension axioms

We introduce two axioms—Upward Extension* and Downward Extension*—that complement UE and DE by allowing *partial* attachments, compensated by a requirement that the appended hierarchy be (weakly) more hierarchical than the base.

Partial-append compound (set-valued operator). For $h, h_o \in \mathscr{H}$, let $h \uplus h_o$ denote the *set* of hierarchies obtained by:

1. taking the disjoint union of h and h_o, and
2. adding one or more edges from *at least one* leaf of h to *at least one* root of h_o.

Only leaf-to-root cross-edges are allowed; no other cross-edges are permitted. Note that

attachments may be *partial*: not every leaf of h need connect, and not every root of h_o need be attached.

Remark 5. As with the complete operators, $h \uplus h_o \neq \emptyset$ for all $h, h_o \in \mathscr{H}$ (e.g., one can pick any leaf ℓ of h and any root r of h_o and add the edge (ℓ, r)). The operator $\uplus$ is *set-valued* and *order-dependent* (non-commutative). Moreover,

$$h_o \oplus h \;\subseteq\; h_o \uplus h \quad \text{and} \quad h \boxplus h_o \;\subseteq\; h \uplus h_o,$$

with strict inclusions whenever the completeness conditions of $\oplus$ or $\boxplus$ fail (e.g., incomplete leaf coverage or non-maximum-depth attachments).

This operation is illustrated in Figure 7 (where an incomplete, non-maximum-depth attachment yields $h' \in h \uplus h_o$ but $h' \notin h \oplus h_o$ and $h' \notin h \boxplus h_o$) and Figure 10 (another instance of $h' \in h \uplus h_o$).

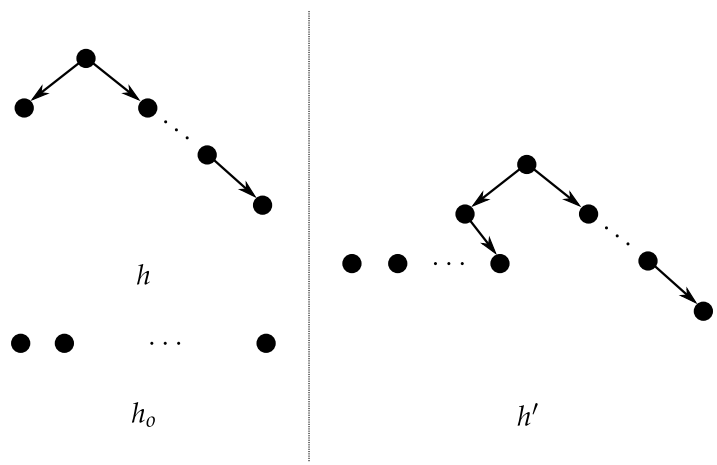

Figure 10: $h' \in h \uplus h_o$.

We now formulate the partial extension axioms in the index-first style.

Upward Extension* (UE*). A hierarchical index $I : \mathscr{H} \to \mathbb{R}$ satisfies UE* if for all $h_o, h \in \mathscr{H}$ and every $h^* \in h_o \uplus h$, $I(h_o) \geq I(h)$ implies $I(h^*) > I(h)$.

Downward Extension* (DE*). A hierarchical index $I : \mathscr{H} \to \mathbb{R}$ satisfies DE* if for all $h, h_o \in \mathscr{H}$ and every $h^* \in h \uplus h_o$, $I(h_o) \geq I(h)$ implies $I(h^*) > I(h)$.

Conditional monotonicity. UE* and DE* are conditional, self-referential axioms: they restrict I's behavior only when I itself ranks the appendage weakly above the base (i.e., $I(h_o) \geq I(h)$). The dominance premise is uniform: it always compares the appended hierarchy h_o to the base h, independent of whether h_o is appended above (via UE*) or below (via DE*).

Interpreting UE* versus UE. UE requires a *complete* upward attachment (every leaf of h_o connects to roots of h), with *no* restriction on the hierarchical status of h_o. In contrast, UE* allows *partial* upward attachment through $\uplus$, provided that the appended hierarchy h_o is itself (weakly) more hierarchical than the base h. Intuitively, a sufficiently hierarchical structure on top can raise overall hierarchy even if it is only partially attached.

Interpreting DE* versus DE. DE requires *complete* downward attachment at the boundary of h (from maximum-depth leaves of h to *every* root of h_o). DE* relaxes both requirements via $\uplus$: only *partial* attachment is necessary, and maximum-depth targeting is not required—again compensated by the premise $I(h_o) \geq I(h)$.

4 Hierarchical indices

We now examine hierarchical indices and their relationship to the axioms introduced in the preceding section.

Our analysis yields three distinct families of hierarchical indices, corresponding to three competing paradigms for what "hierarchy" should measure: the *average* (intensive) family, the *total* (extensive) family, and the *concentration* (density) family. We provide an axiomatic characterization of each class, highlighting the trade-offs implied by the separability and scaling axioms. A re-evaluation of established hierarchical measures from the literature through the lens of our axiomatic framework is deferred to Section 5.

Hierarchical indices offer computational advantages through simple formulae, avoiding the complexities and incompleteness of pre-orders. This makes them readily applicable to real-world datasets. However, by inducing a complete order on hierarchies, these indices necessarily make definitive comparisons between any two hierarchical structures—even in ambiguous cases where the structural axioms alone would not single out a more hierarchical structure.

4.1 Classical separability: AS, ADD, and QS

In classical measurement theory, separability dictates that an aggregate index can be cleanly decomposed into a sum of independent, localized evaluations. The three separability axioms introduced in this section adapt this classical paradigm to hierarchical networks by restricting the global evaluation to an additive aggregation of localized environment functions.

These axioms share an additive numerator but differ only in normalization, yielding three distinct metrics: Additivity (ADD) evaluates total structural mass, Additive Separability (AS) measures per-capita experience, and Quadratic Separability (QS) assesses global structural density. Consequently, all three index families rest on the same classical foundation of additive decomposition, distinguished solely by their scaling with organizational size.

Additive Separability and Additivity are standard index properties used extensively in the literature on inequality and poverty, while quadratic normalizations underpin classical measures of relative concentration.

To translate the concept of separability from scalar distributions (like income) to structural networks, we must first isolate the structural information relevant to an individual's position. For any hierarchy $h \in \mathscr{H}$ and any node $v \in V(h)$, we define the *hierarchical environment* of v, denoted $N(v, h)$, as the subgraph of h induced by v and all nodes reachable from or to v via directed paths (i.e., the entirety of v's direct and indirect supervisors and subordinates). Let $\mathscr{N}$ denote the space of all such valid hierarchical environments.

Additive Separability (AS). A hierarchical index $I : \mathscr{H} \to \mathbb{R}$ satisfies AS if there exists an environment evaluation function $f : \mathscr{N} \to \mathbb{R}$ such that for any hierarchy $h \in \mathscr{H}$,

$$I(h) = \frac{1}{|V(h)|} \sum_{v \in V(h)} f\big(N(v, h)\big).$$

The intuition behind the AS axiom is rooted in the micro-foundations of organizational structure. Instead of evaluating a hierarchy purely as a unitary, macro-level shape, AS mandates that the overall hierarchical degree of an organization is the population average of the individual hierarchical "scores" of its constituent members.

By restricting the domain of the evaluation function f to the localized environment $N(v, h)$, AS explicitly enforces structural independence. This mirrors the implicit independence embedded in the FGT poverty indices (Foster et al., 1984), where an individual's contribution depends solely on their own localized state, independent of the broader distribution. In our network context, this localized evaluation guarantees subgroup consistency: if a

specific branch of an organization undergoes restructuring that increases its local hierarchical depth while the remainder of the organization is untouched, the structural evaluations of disconnected employees remain mathematically unchanged, and therefore leaves other nodes' evaluations unchanged; if the restructuring raises the evaluations $f(N(v,h))$ for at least one node and lowers none, then the AS index strictly increases. Furthermore, AS serves as a natural strengthening of IRP. Because AS calculates a population average, scaling an organization by constructing k exact replicas simply multiplies both the numerator (the total sum of node evaluations) and the denominator (the total population) by k, leaving the overall average identical.

Additivity (ADD). A hierarchical index $I : \mathscr{H} \to \mathbb{R}$ satisfies ADD if there exists an environment evaluation function $f : \mathscr{N} \to \mathbb{R}$ such that for any hierarchy $h \in \mathscr{H}$,

$$I(h) = \sum_{v \in V(h)} f\big(N(v,h)\big).$$

In the classical measurement literature on inequality and poverty (e.g., Shorrocks, 1980), an index is strictly additive if the aggregate measure is the simple sum of individual evaluations. ADD serves as the direct network-theoretic analog to this classical strict additivity. In a network context, an individual's "endowment" is not a scalar quantity, but their localized structural position, $N(v,h)$. ADD dictates that the overall hierarchical degree of the system is the absolute sum of these individual hierarchical burdens, conceptualizing hierarchy as an *extensive* property—a cumulative, absolute volume of organizational structure—rather than an *intensive*, per-capita average.

This extensive formulation mathematically implies standard additivity over disjoint organizational components: for any composite hierarchy composed of independent sub-hierarchies h_1 and h_2, it follows that $I(h_1 \cup h_2) = I(h_1) + I(h_2)$. However, unlike standard component-level additivity, the node-level formulation of ADD ensures that structural separability holds internally *within* connected organizational structures as well.

Under this paradigm, every supervisory relationship contributes a positive, absolute amount of structure to the world. Consequently, creating an exact replica of an existing organization does not leave the overall hierarchical degree unchanged; it doubles it, reflecting the fact that twice as many subordination relationships now exist. Therefore, any index satisfying ADD guarantees ERP. Furthermore, ADD protects the index from the "dilution effect" inherent to population-averaged indices. Because the index does not rely on a population denominator, appending or evaluating a large, completely flat organization

alongside a highly vertical one will never drag the overall hierarchical score downward; independent flat structures simply contribute zero additional hierarchy under a natural normalization ensuring that $I(h) = 0$ whenever h consists of a single isolated node (cf. INN below).

This fundamental dichotomy between additive (extensive) and decomposable (intensive) aggregation is well-established in axiomatic index theory. It mirrors the distinction between the absolute *total poverty gap*—which strictly sums the deprivations of all individuals to capture the total societal burden—and normalized, population-averaged measures like the standard FGT index. Similarly, in population ethics, pure Additivity characterizes Classical Utilitarianism, whereas decomposability characterizes Average Utilitarianism (Blackorby et al., 2005).

Yet this binary leaves a third structural perspective unaddressed: the view of hierarchy as a measure of *global concentration* or *network density*. In many organizational contexts, the relevant question is not how much hierarchy exists in absolute terms, nor what the average individual experiences, but how densely packed the authority structure is relative to the population's theoretical capacity for subordination. To capture this density-based intuition, we introduce a third separability paradigm that replaces linear or absent scaling with a quadratic denominator. This formulation effectively measures hierarchy as the ratio of realized supervisory relationships to maximum potential dyadic structure, explicitly penalizing fragmentation as organizational size increases.

Quadratic Separability (QS). A hierarchical index $I : \mathscr{H} \to \mathbb{R}$ satisfies QS if there exists an environment evaluation function $f : \mathscr{N} \to \mathbb{R}$ such that for any hierarchy $h \in \mathscr{H}$,

$$I(h) = \frac{1}{|V(h)|^2} \sum_{v \in V(h)} f\big(N(v, h)\big).$$

While AS and ADD measure local environments against a linear population and an absolute baseline, respectively, QS introduces a quadratic penalty. In classical economics, relative concentration and inequality are frequently measured by evaluating differences across all possible *pairs* of individuals. The canonical Gini coefficient, for instance, normalizes its sum of pairwise absolute differences by the square of the total population size to capture the density of inequality across the entire social fabric. QS adapts this dyadic normalization to organizational structure. Because hierarchical chains are fundamentally built upon ordered pairs of individuals (boss and subordinate), dividing the sum of individual evaluations by $|V(h)|^2$ conceptualizes hierarchy as the ratio of *realized* localized structure to *maximum*

potential dyadic structure.

Just as the Gini coefficient is not strictly additively decomposable due to cross-group interaction terms, component-level aggregation under QS penalizes structural fragmentation. For a composite hierarchy of independent sub-hierarchies h_1 and h_2, QS implies an aggregation where each component's score is weighted by the square of its proportional share of the total population. The "missing" weight in this convex combination can be interpreted as the potential cross-component relationships that fail to exist because the groups are isolated.

By aggressively penalizing structural voids, QS implies CRP whenever $I(h) > 0$ for non-flat h. Replicating an organization duplicates the numerator (actual structure) but scales the denominator by the square of the replication factor (potential structure), strictly diluting the global hierarchical density. Thus, replacing the denominator defines the theoretical lens: dividing by $|V(h)|^0$ measures cumulative mass (ERP), dividing by $|V(h)|^1$ measures the per-capita average (IRP), and dividing by $|V(h)|^2$ measures global concentration (CRP).

Remark 6 (scaling exponents and higher-order structures)**.** The choice of the exponent p in the normalization denominator $|V(h)|^p$ dictates the relational order against which the network is being evaluated. Dividing by $|V(h)|^2$ under QS is theoretically grounded in the fact that hierarchies are fundamentally constructed from binary relations (boss-subordinate pairs). Because the theoretical maximum number of such directed pairs in any network scales quadratically, normalizing by $p = 2$ correctly bounds the measure as a true *dyadic density*.

Extending this logic, dividing by higher integer powers evaluates the concentration of higher-order relational structures. For example, $p = 3$ would normalize against the theoretical maximum number of node triads, measuring the density of three-tier chains ($i \to j \to k$) rather than simple pairs. A generic integer power p evaluates the network's saturation relative to p-tuple subgraphs.

Conversely, fractional exponents (such as $p = 1.5$ or $p = 2.5$) correspond to empirical scaling regimes where structural capacity grows with organization size at a non-integer rate. In dynamic network analysis, empirical graphs frequently follow a "densification power law" (Leskovec et al., 2005) wherein edges proliferate super-linearly but remain strictly sub-quadratic as nodes are added. Translating this to organizational measurement, a denominator like $|V(h)|^{1.5}$ would benchmark structural density against empirically observed scaling boundaries rather than absolute mathematical maxima, potentially accommodating sparse-but-growing reporting architectures that outpace linear growth without approaching

full dyadic saturation. Such sub-quadratic scaling may reflect cognitive limits on relationship management (Dunbar, 1992) or administrative constraints on managerial span-of-control (Williamson, 1967), which jointly limit the density of feasible supervisory relationships even as organizations expand.

4.2 Impossibility

We first observe that the normalized separability axioms (AS and QS) are incompatible with the strict structural extension axioms (UE and DE).

Throughout this section, we restrict our attention to *regular* separable indices. A separable index (i.e., one satisfying AS, ADD, or QS, and thus admitting a local evaluation function f) is *regular* if it is strictly non-negative ($I(h) \geq 0$ for all h, and $I(h) > 0$ for complex non-flat structures) and its environment evaluation function does not asymptotically saturate (for any $K > 0$, there exists an environment N such that $f(N) > K$). Finally, a regular separable index exhibits monotonically bounded marginal returns: expanding a local environment cannot decrease its hierarchical evaluation, and its growth is strictly bounded. Formally, there exists a finite constant $M > 0$ such that for any hierarchical environments $N \subseteq N'$ (where $\subseteq$ denotes induced-subgraph inclusion), the change in evaluation satisfies

$$0 \leq f(N') - f(N) \leq M(|V(N')| - |V(N)|).$$

Theorem 1. *There exists no regular hierarchical index satisfying* A *that simultaneously satisfies either* AS *or* QS *and either* UE *or* DE.

Proof. Suppose, for contradiction, that there exists a regular hierarchical index $I : \mathscr{H} \to \mathbb{R}$ satisfying A, either AS or QS, and also UE. Because I satisfies AS or QS, there exist $f : \mathscr{N} \to \mathbb{R}$ and $p \in \{1, 2\}$ such that for every $h \in \mathscr{H}$,

$$I(h) = \frac{1}{|V(h)|^p} \sum_{v \in V(h)} f\big(N(v, h)\big). \tag{1}$$

Let $c_2 = f(N_2)$ be the evaluation of a 2-node directed chain.

Step 1 (construct a baseline hierarchy h with $I(h) = C$ large enough). *Case $p = 1$ (AS).* Because f does not saturate, pick an environment N^* with $f(N^*) > c_2 + M$. Let $K = f(N^*)$, so $K > c_2 + M$. Form h_1 by appending a flat hierarchy of k nodes upward

to N^*. Then for each of the k new nodes u, we have $N^* \subseteq N(u, h_1)$. By monotonicity, $f(N(u, h_1)) \geq f(N^*) = K$. Hence

$$\sum_{v \in V(h_1)} f(N(v, h_1)) \geq kK,$$

so

$$I(h_1) = \frac{1}{|V(h_1)|} \sum_{v \in V(h_1)} f(N(v, h_1)) \geq \frac{kK}{|V(h_1)|}.$$

As k increases, the fraction of nodes among $V(h_1)$ with evaluation at least K tends to 1, so we can choose k large enough so that

$$I(h_1) \geq c_2 + M + \delta$$

for some fixed $\delta > 0$.

Write $s = |V(N^*)|$, so $|V(h_1)| = m = k + s$. For each of the k new nodes u, the environment $N(u, h_1)$ is obtained from N^* by adding the single node u as a new ancestor. By bounded marginal returns,

$$f(N(u, h_1)) \leq f(N^*) + M = K + M.$$

For each node $w \in V(N^*)$, the construction adds at most k new ancestors to w, so again by bounded marginal returns,

$$f(N(w, h_1)) \leq f(N(w, N^*)) + Mk.$$

Let

$$A^* = \sum_{w \in V(N^*)} f(N(w, N^*)),$$

which is a finite constant independent of k. Summing the above inequalities yields

$$\sum_{v \in V(h_1)} f(N(v, h_1)) \leq k(K + M) + A^* + sMk.$$

Dividing by $m = k + s$ gives

$$I(h_1) \leq \frac{k(K + M + sM)}{k + s} + \frac{A^*}{k + s} \leq (K + M + sM) + A^* = \overline{B},$$

where $\overline{B}$ is a constant independent of k (hence independent of m).

Case $p = 2$ (QS). Choose any complex non-flat hierarchy h_1. Regularity implies $I(h_1) > 0$.

Now let h be the disjoint union of h_1 and one isolated node r_2. Write $m = |V(h_1)|$, so $|V(h)| = m + 1$. Define $C = I(h)$.

If $p = 2$, then $C > 0$ because h is complex and regularity gives strict positivity.

If $p = 1$, then using (1) and letting $c_1 = f(N_1)$ (the one-node environment),

$$C = I(h) = \frac{mI(h_1) + c_1}{m+1}.$$

Hence

$$C \geq I(h_1) - \frac{I(h_1) - c_1}{m+1}.$$

By the uniform bound $I(h_1) \leq \overline{B}$, we have

$$\frac{I(h_1) - c_1}{m+1} \leq \frac{\overline{B} - c_1}{m+1}.$$

Since $m = k + s$ can be made arbitrarily large by increasing k, we may choose k (and thus m) large enough so that

$$\frac{\overline{B} - c_1}{m+1} \leq \frac{\delta}{2},$$

which implies

$$\frac{I(h_1) - c_1}{m+1} \leq \frac{\delta}{2}.$$

Then

$$C \geq c_2 + M + \frac{\delta}{2} > c_2 + M.$$

Step 2 (append a large flat hierarchy above h in a way that dilutes the index). Let h_o be a completely flat hierarchy of n nodes. Form a compound $h^* \in h_o \oplus h$ as follows: choose exactly one node $v^* \in V(h_o)$ and connect v^* to all roots of h_1 and to r_2. Connect each of the remaining $n - 1$ nodes of h_o exclusively to r_2 (Figure 11).

Let

$$S(h) = \sum_{v \in V(h)} f(N(v, h)),$$

which is a constant independent of n.

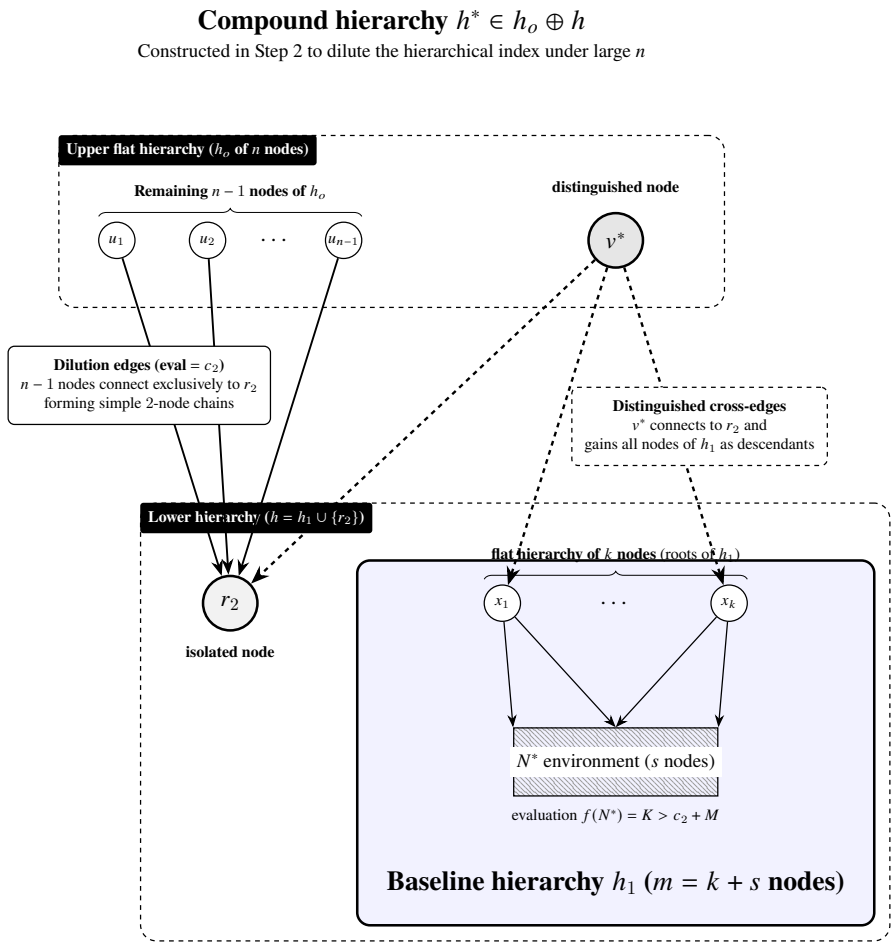

Figure 11: Compound hierarchy h^*.

We now bound

$$S(h^*) = \sum_{v \in V(h^*)} f(N(v, h^*))$$

from above by an affine function of n with an explicit constant term.

(i) Nodes in h_1. Each of the m nodes of h_1 gains exactly one new ancestor (v^*). By bounded marginal returns,

$$f(N(v, h^*)) - f(N(v, h)) \leq M \quad \text{for each } v \in V(h_1).$$

Therefore the total increase contributed by nodes in h_1 is at most mM.

(ii) The node r_2. The node r_2 gains exactly n new ancestors (all nodes of h_o). Thus, by bounded marginal returns,

$$f(N(r_2, h^*)) - f(N(r_2, h)) \leq Mn.$$

(iii) The $n-1$ nodes in $h_o \setminus \{v^*\}$. Each such node becomes the root of a simple 2-node chain pointing to r_2. By anonymity, their local environment evaluations equal c_2. Hence their total contribution is exactly $(n-1)c_2$.

(iv) The single node v^*. The node v^* has an environment of size $m+2$ (itself, r_2, and all nodes of h_1). This environment does not depend on n. Let

$$B = f(N(v^*, h^*)),$$

which is a finite constant independent of n.

Combining (i)–(iv) yields

$$S(h^*) \leq S(h) + mM + Mn + (n-1)c_2 + B = (c_2 + M)n + \underbrace{\big(S(h) + mM + B - c_2\big)}_{K_0},$$

where K_0 is a constant independent of n.

Therefore, using (1) and $|V(h^*)| = n + m + 1$,

$$I(h^*) = \frac{S(h^*)}{(n+m+1)^p} \leq \frac{(c_2 + M)n + K_0}{(n+m+1)^p}. \tag{2}$$

Step 3 (take n large and contradict UE). *Case $p = 1$ (AS).* From (2),

$$I(h^*) \le (c_2 + M)\frac{n}{n+m+1} + \frac{K_0}{n+m+1}.$$

Letting $n \to \infty$ gives

$$\limsup_{n\to\infty} I(h^*) \le c_2 + M.$$

But we built h so that $I(h) = C > c_2 + M$. Hence there exists n_0 such that for all $n \ge n_0$, $I(h^*) < C = I(h)$, contradicting UE (which requires $I(h^*) > I(h)$ for every $h^* \in h_o \oplus h$).

Case $p = 2$ (QS). From (2),

$$I(h^*) \le \frac{(c_2 + M)n + K_0}{(n+m+1)^2} \le \frac{c_2 + M}{n+m+1} + \frac{K_0}{(n+m+1)^2}.$$

Letting $n \to \infty$ gives $\lim_{n\to\infty} I(h^*) = 0$. But we built h so that $I(h) = C > 0$. Thus for large n, $I(h^*) < C = I(h)$, again contradicting UE.

Step 4 (the same contradiction for DE). The argument for DE is symmetric. Let h be the disjoint union of a massive branch h_1 and a single isolated leaf l_2. Append a flat hierarchy of n new nodes *below* by forming $h^* \in h \boxplus h_{\text{flat}}$ so that every new node attaches exclusively to l_2. Then the same bounded-marginal-returns calculation shows that the numerator $S(h^*)$ is at most $(c_2 + M)n + K_0'$ for some constant K_0', while the denominator is $(n+m+1)^p$. Hence $I(h^*)$ falls below $I(h)$ for large n, contradicting DE.

This completes the contradiction and proves the theorem. ■

We further establish that while the population-averaging of AS permits compliance with relaxed extension axioms, the quadratic normalization of QS fundamentally precludes even partial structural extensions.

Theorem 2. *There exists no regular hierarchical index satisfying A and QS that simultaneously satisfies the partial extension axioms UE* or DE*.*

Proof. Suppose, by way of contradiction, that there exists a regular hierarchical index $I : \mathscr{H} \to \mathbb{R}$ that satisfies A, QS, and UE*. (The proof for DE* is symmetric.)

Because I satisfies QS, there exists an environment evaluation function $f : \mathscr{N} \to \mathbb{R}$ such that for every $h \in \mathscr{H}$,

$$I(h) = \frac{1}{|V(h)|^2} \sum_{v \in V(h)} f\big(N(v,h)\big).$$

Let $c_1 = f(N_1)$ be the evaluation of the one-node environment, and let $c_2 = f(N_2)$ be the evaluation of a 2-node directed chain.

Construct a baseline hierarchy $h \in \mathscr{H}$ as the disjoint union of a complex (non-flat) hierarchy h_{core} and a single isolated node v_{flat}. Let $N = |V(h)|$. Define the aggregate evaluation (numerator)

$$S(h) = \sum_{v \in V(h)} f\big(N(v, h)\big),$$

so that by QS we have

$$I(h) = \frac{S(h)}{N^2}.$$

Because h is complex, regularity implies $I(h) = C > 0$.

Let h_o be an exact isomorphic copy of h. By A, $I(h_o) = I(h) = C$, so the weak dominance premise in UE* is satisfied.

Form a compound $h^* \in h_o \uplus h$ by adding exactly one directed edge from the flat node in h_o to the flat node in h, and adding no other cross-edge.

Claim A. *The local environments (and hence the evaluations) of the* $2N-2$ *nodes belonging to the two core components remain unchanged when passing from the disjoint union* $h_o \cup h$ *to* h^*.

Proof of Claim A. Indeed, in $h_o \cup h$ the two flat nodes are isolated and have no edges to any core node. The single added edge connects the two flat nodes to each other but still creates no directed path from any core node to either flat node, nor from either flat node to any core node. Hence for every core node v in either copy, its set of ancestors and descendants is identical in $h_o \cup h$ and in h^*, so $N(v, h^*) = N(v, h_o \cup h)$ and thus $f(N(v, h^*)) = f(N(v, h_o \cup h))$. □

Claim B. *The only two nodes whose environments change are the two flat nodes, and each of them moves from environment* N_1 *to environment* N_2.

Proof of Claim B. Before the linkage, each flat node is isolated, so its environment is N_1. After adding the single edge between them, the induced environment of each flat node is exactly the 2-node chain N_2 (each flat node can reach or be reached by the other). Therefore, each flat node's evaluation changes from c_1 to c_2, so the combined change in the numerator is

$$\big(c_2 - c_1\big) + \big(c_2 - c_1\big) = 2(c_2 - c_1).$$

Moreover, by bounded marginal returns (the Lipschitz bound in regularity), adding one

node to an environment changes its evaluation by at most M, hence $0 \le c_2 - c_1 \le M$, and therefore $0 \le 2(c_2 - c_1) \le 2M$. □

Combining Claim A and Claim B yields an exact expression for the numerator of h^*. Let $S(h_o \cup h)$ denote the numerator of the disjoint union $h_o \cup h$. Since h_o is isomorphic to h, we have $S(h_o \cup h) = S(h_o) + S(h) = 2S(h) = 2N^2C$. After the linkage, the only change is $2(c_2 - c_1)$, hence

$$S(h^*) = 2S(h) + 2(c_2 - c_1) = 2N^2C + 2(c_2 - c_1).$$

Since $|V(h^*)| = |V(h_o \cup h)| = 2N$, the QS formula gives

$$I(h^*) = \frac{S(h^*)}{(2N)^2} = \frac{2N^2C + 2(c_2 - c_1)}{4N^2} = \frac{C}{2} + \frac{c_2 - c_1}{2N^2}.$$

Using $0 \le c_2 - c_1 \le M$, we obtain the explicit bound

$$I(h^*) \le \frac{C}{2} + \frac{M}{2N^2}.$$

Because $C > 0$, choose N large enough so that

$$\frac{M}{2N^2} < \frac{C}{2}.$$

For such an N we have

$$I(h^*) \le \frac{C}{2} + \frac{M}{2N^2} < \frac{C}{2} + \frac{C}{2} = C = I(h).$$

Thus $I(h^*) < I(h)$, which contradicts the strict increase conclusion required by UE*.

The proof for DE* is symmetric: take the same two copies and add a single admissible cross-edge witnessing $h^* \in h \uplus h_o$ in the direction relevant for DE*; the same numerator computation yields $I(h^*) < I(h)$ for sufficiently large N, contradicting DE*. ■

Intuition (Theorem 1, Theorem 2). The impossibilities have distinct structural sources.

Theorem 1 uses a *dilution* argument. Under AS or QS, the index is a normalized average of node-level environment scores. The strict extension axioms UE/DE require that every admissible vertical attachment strictly raises the index. But the appending operators allow one to add arbitrarily many new nodes whose local environments remain topologically

shallow because reachability is routed through a single screening root/leaf. Those nodes contribute only uniformly bounded scores, so as their number grows they dominate the normalization and pull the index down (for QS, the normalization is quadratic, so linear structural additions are overwhelmed).

Theorem 2 uses a *concentration–fragmentation* argument specific to QS. The partial axioms UE*/DE* block the most direct dilution move, so the construction uses two equally hierarchical copies connected by a minimal admissible link. Under quadratic normalization, duplicating the population expands the denominator by a factor of 4, while a single admissible cross-edge affects only the local environments of two nodes. This small local change cannot compensate for the massive fragmentation created by having two large cores with almost no cross-links, so the QS score falls, violating UE*/DE*.

4.3 A class of bilateral average indices

Theorem 1 establishes a fundamental limit on hierarchical measurement: no regular index can simultaneously evaluate hierarchy as a localized average (AS) and remain strictly monotonic to the unrestricted addition of vertical layers (UE/DE). To proceed, an analyst must choose which structural property to relax.

Before characterizing our primary index families, we introduce a normative structural principle regarding internal organizational routing. In many contexts, the hierarchical burden of an organization is driven by the pure reachability of authority rather than the specific, intermediate chains of command. Because AS explicitly decomposes the index into a sum of localized evaluations, we can unambiguously define this topological neutrality based directly on these individual structural contributions.

Condition 1 (Environment Topology Neutrality, ETN)**.** *Let $I : \mathscr{H} \to \mathbb{R}$ be a hierarchical index satisfying one of the separability axioms AS, ADD, or QS with environment evaluation function $f : \mathscr{N} \to \mathbb{R}$. We say that I satisfies ETN if for any hierarchical environment $N \in \mathscr{N}$ centered at a distinguished node v, the local evaluation $f(N)$ is invariant to any restructuring of the edges in N that yields a valid hierarchical environment centered at v, provided the sets of v's direct and indirect supervisors and v's direct and indirect subordinates remain unchanged.*

The intuition behind ETN is that an individual's hierarchical burden is determined by *who* supervises them and *who* they supervise—not how those supervisors and subordinates relate to one another internally. This encompasses several distinct structural dimensions:

the internal reporting arrangements among a node's supervisors, the internal arrangements among its subordinates, cross-connections that bypass intermediate managers entirely, and whether a given reporting relationship is direct or indirect. ETN asserts that none of these internal topological variations affect the node's local hierarchical score, provided the absolute sets of superiors and subordinates remain perfectly intact.

ETN carries genuine normative content in the directed acyclic graph setting. Unlike trees, a DAG permits edges that bypass an intermediate node entirely, and it permits supervisors or subordinates to be connected through chains of varying length and density. ETN rules out sensitivity to all such structural variation simultaneously, in a single coherent normative statement.

When ETN is combined with the population-average structure of AS and the node-label independence of Anonymity (A), it constrains the functional form of an index, forcing it to evaluate organizations exclusively through the lens of individual depth and span. We formalize this structural reduction below.

Lemma 1. *A hierarchical index $I : \mathscr{H} \to \mathbb{R}$ satisfies A, AS, and ETN if and only if there exists a function $g : \mathbb{Z}_+ \times \mathbb{Z}_+ \to \mathbb{R}$ such that for every $h \in \mathscr{H}$*

$$I(h) = \frac{1}{|V(h)|} \sum_{v \in V(h)} g\big(s_h(v), \sigma_h(v)\big).$$

Proof. **Sufficiency (⇐).** Suppose

$$I(h) = \frac{1}{|V(h)|} \sum_{v \in V(h)} g(s_h(v), \sigma_h(v))$$

for some function g. Define the local environment evaluation as $f(N(v, h)) = g(s_h(v), \sigma_h(v))$. Because the cardinalities $s_h(v)$ and $\sigma_h(v)$ are strictly invariant under graph isomorphism, I is unchanged by any relabeling of the nodes, ensuring A holds. The additive functional form directly satisfies AS. Furthermore, because g depends exclusively on the sizes of the ancestor and descendant sets, any internal restructuring of edges that preserves those sets leaves the local evaluation unchanged, satisfying ETN.

Necessity (⇒). Assume I satisfies A, AS, and ETN.

Step 1. Symmetrization. By AS, there exists a local evaluation function $f : \mathcal{N} \to \mathbb{R}$ such that for any hierarchy $h \in \mathcal{H}$,

$$I(h) = \frac{1}{|V(h)|} \sum_{v \in V(h)} f(N(v, h)).$$

Although f itself is not assumed to be label-invariant, the global index I is. Let Π denote the set of all node-label permutations (relabelings) of the hierarchy h. By A, $I(h)$ is invariant under any $\pi \in \Pi$. Averaging the AS representation over all such permutations yields:

$$I(h) = \frac{1}{|\Pi|} \sum_{\pi \in \Pi} I(\pi(h)) = \frac{1}{|V(h)|} \sum_{v \in V(h)} \underbrace{\left[\frac{1}{|\Pi|} \sum_{\pi \in \Pi} f(\pi(N(v, h))) \right]}_{\bar{f}(N(v,h))}. \tag{3}$$

The bracketed term defines a symmetrized local evaluation function $\bar{f}$. Note that for any relabeling π, we have $\pi(N(v, h)) = N(\pi(v), \pi(h))$; thus the distinguished center node is transported along with the relabeling.

Claim 3. *The symmetrized evaluation $\bar{f}(N(v, h))$ depends strictly on the rooted isomorphism class (isomorphisms mapping the center to the center) of $N(v, h)$.*

Proof of Claim 3. To formally verify that the inner average in (3) depends only on the local environment, consider how a global permutation $\pi \in \Pi$ acts upon the environment's nodes $V(N(v, h))$. Let T be the set of all valid label injections $\tau : V(N(v, h)) \to V(h)$. We can partition the set of all global permutations Π into disjoint subsets $\Pi_\tau = \{\pi \in \Pi \mid \pi|_{V(N(v,h))} = \tau\}$, grouping together all permutations that apply the exact same labels to the local environment.

Because the function f evaluates only the isolated environment, its output depends solely on τ and is entirely blind to how π maps the remaining nodes $V(h) \setminus V(N(v, h))$. Thus, for any $\pi \in \Pi_\tau$, we have $f(\pi(N(v, h))) = f(\tau(N(v, h)))$.

The number of global permutations in each subset Π_τ is exactly $|\Pi_\tau| = (|V(h)| - |V(N(v, h))|)!$, representing all possible ways to distribute the leftover labels to the nodes

outside the environment. We can therefore expand and factor the sum explicitly:

$$\begin{aligned}
\frac{1}{|V(h)|!}\sum_{\pi\in\Pi} f(\pi(N(v,h))) &= \frac{1}{|V(h)|!}\sum_{\tau\in T}\sum_{\pi\in\Pi_\tau} f(\pi(N(v,h))) \\
&= \frac{1}{|V(h)|!}\sum_{\tau\in T}\sum_{\pi\in\Pi_\tau} f(\tau(N(v,h))) \\
&= \frac{1}{|V(h)|!}\sum_{\tau\in T}|\Pi_\tau| f(\tau(N(v,h))) \\
&= \frac{(|V(h)|-|V(N(v,h))|)!}{|V(h)|!}\sum_{\tau\in T} f(\tau(N(v,h))).
\end{aligned}$$

Because the fraction on the right is the exact reciprocal of the total number of valid injections $|T|$, the ambient hierarchy merely contributes a combinatorial multiplier that is perfectly neutralized. The operation mathematically reduces to an exact, uniform arithmetic mean over all possible label injections of the isolated environment. If two environments are rooted isomorphic, summing over all possible injections from $V(h)$ will generate the exact same set of identically-evaluated terms. By integrating out the specific original node identities in this manner, $\bar{f}$ yields the exact same value for any two environments sharing the same topology, making it a well-defined function evaluated strictly on rooted isomorphism classes. □

We thus have

$$I(h) = \frac{1}{|V(h)|}\sum_{v\in V(h)} \bar{f}(N(v,h)).$$

Step 2. Canonical reduction via ETN. We first explicitly verify that because ETN holds for the witness f, it is inherited by the symmetrized witness $\bar{f}$. Let N and N' be two environments centered at v that differ only by an ETN-admissible rewiring. By the definition of ETN, any relabeling π preserves the cardinalities of the ancestor and descendant sets, meaning the relabeled environments $\pi(N)$ and $\pi(N')$ also differ only by an ETN-admissible rewiring. Since the original witness f satisfies ETN, it follows that $f(\pi(N)) = f(\pi(N'))$ for every possible relabeling $\pi \in \Pi$. Summing these identical evaluations over all $\pi \in \Pi$ and dividing by $|\Pi|$ guarantees exactly $\bar{f}(N) = \bar{f}(N')$.

Fix any node $v \in V(h)$. Notice that ETN restricts the evaluation based *only* on the absolute sets of ancestors and descendants of the centered node v. It is strong enough to allow us to completely ignore and eliminate all other internal topology (i.e., edges strictly

among ancestors, strictly among descendants, or directly from ancestors to descendants). We restructure $N(v, h)$ into a canonical "double-star" environment, denoted $D_{s,\sigma}$, which consists of exactly $s = s_h(v)$ independent roots pointing directly to v, and v pointing directly to exactly $\sigma = \sigma_h(v)$ independent leaves, with no other internal edges.

Such a rewiring always exists: delete all internal edges among ancestors, among descendants, and between ancestors and descendants; then add directed edges $u \to v$ for each ancestor u, and $v \to w$ for each descendant w. The resulting environment is acyclic and preserves exactly the absolute sets of ancestors and descendants of v. By ETN, this structural rewiring yields:

$$\bar{f}(N(v, h)) = \bar{f}(D_{s_h(v),\sigma_h(v)}).$$

Step 3. Defining the universal function. Because $\bar{f}$ is strictly invariant on rooted isomorphism classes (from Claim 1) and the double-star $D_{s,\sigma}$ is completely characterized up to rooted isomorphism by the cardinalities s and σ, the evaluation $\bar{f}(D_{s,\sigma})$ depends strictly on these two integers. We can therefore define a universal function $g : \mathbb{Z}_+ \times \mathbb{Z}_+ \to \mathbb{R}$ as:

$$g(s, \sigma) = \bar{f}(D_{s,\sigma}).$$

Substituting this directly into the symmetrized AS representation from Step 1 gives:

$$I(h) = \frac{1}{|V(h)|} \sum_{v \in V(h)} g\big(s_h(v), \sigma_h(v)\big).$$

This completes the proof. ■

4.3.1 Intensive Population Growth (IPG)

Definition 2. A hierarchical index $I : \mathscr{H} \to \mathbb{R}$ is a *bilateral average index* if there exists a local evaluation function $g : \mathbb{Z}_+ \times \mathbb{Z}_+ \to \mathbb{R}$ such that for all $h \in \mathscr{H}$,

$$I(h) = \frac{1}{|V(h)|} \sum_{v \in V(h)} g\big(s_h(v), \sigma_h(v)\big).$$

To capture the requirement that significant systemic structural shifts must leave a persistent trace on the overall organizational evaluation, we now introduce a regularity condition specifically for bilateral average indices. Formally, let I be a bilateral average

index characterized by a local evaluation function g. Fix a natural identification of each node in the two disjoint copies of h_k with its counterpart in the compound $h'_k \in h_k \uplus h_k$. Write $\Delta s_k(v)$ and $\Delta\sigma_k(v)$ for the changes in depth and span of a node v between its baseline copy and its position in h'_k.

We say that a sequence of compounds $(h'_k \in h_k \uplus h_k)$ is *macroscopic* if a non-vanishing fraction of nodes experience a positive coordinate shift: there exists $\rho > 0$ and k_0 such that for all $k \geq k_0$,

$$\left|\{v \in V(h'_k) : \Delta s_k(v) > 0 \text{ or } \Delta\sigma_k(v) > 0\}\right| \geq \rho|V(h'_k)|.$$

To evaluate these structural shifts, we define the *discrete marginals* of g with respect to depth and span as

$$\Delta_s g(s,\sigma) = g(s+1,\sigma) - g(s,\sigma) \quad \text{and} \quad \Delta_\sigma g(s,\sigma) = g(s,\sigma+1) - g(s,\sigma).$$

To isolate the influence of extreme shifts from a finite set of outlier nodes, let v_0 denote the corresponding node of $v \in V(h'_k)$ in the appropriate baseline copy of h_k. We define the marginal evaluation gain of v as the total change in its evaluation:

$$\Delta g_k(v) = g\big(s_{h'_k}(v), \sigma_{h'_k}(v)\big) - g\big(s_{h_k}(v_0), \sigma_{h_k}(v_0)\big).$$

Letting $\Delta g_k^+(v) = \max\{\Delta g_k(v), 0\}$, we denote by

$$\Delta g^+_{(1)} \geq \Delta g^+_{(2)} \geq \cdots$$

the order statistics of the multiset $\{\Delta g_k^+(v) : v \in V(h'_k)\}$. For any integer $c \geq 1$, we define

$$M_k(c) = \frac{1}{|V(h'_k)|} \sum_{i=1}^{c} \Delta g^+_{(i)}.$$

Intuitively, $M_k(c)$ represents the normalized aggregate structural gain attributable to the top c outlier nodes experiencing the largest positive shifts.

Condition 2 (IPG: Intensive Population Growth)**.** *A bilateral average index I satisfies* IPG *if for any macroscopic sequence of compounds $(h'_k \in h_k \uplus h_k)$ formed from a baseline hierarchy h_k expanding such that $|V(h_k)| \to \infty$, the aggregate structural accumulation strictly outpaces the contributions of any finite set of outlier nodes. Specifically, for any fixed*

integer $c \geq 1$,

$$\liminf_{k\to\infty} \left[I(h'_k) - I(h_k) - M_k(c) \right] > 0.$$

Remark 7. The IPG condition governs how the evaluation function g responds to structural shifts at both the extremes and the aggregate level. If g has discrete marginals bounded from above, the total evaluation gain from any fixed number c of outliers is bounded by a constant, which implies $M_k(c) \to 0$ as the population $|V(h'_k)|$ grows. More importantly, the strict inequality in IPG ensures that macroscopic shifts—where a non-vanishing fraction of the population advances—leave a persistent, strictly positive impact on the overall index. As established in Proposition 1 and Theorem 3, it is precisely this requirement that forces the discrete marginals of g to be strictly bounded away from zero.

Proposition 1. *If a bilateral average index satisfies IPG, its local evaluation function* $g : \mathbb{Z}_+ \times \mathbb{Z}_+ \to \mathbb{R}$ *must be strictly increasing in both arguments and, for any fixed value of one coordinate, have discrete marginals strictly bounded away from zero in the other argument.*

Proof. Suppose I is a bilateral average index characterized by $g(s, \sigma)$. We first prove that g must be strictly increasing in each argument by constructing macroscopic compound sequences that isolate a unit increase in a single coordinate.

Step 1 (strict increase in span). Assume, toward a contradiction, that g is not strictly increasing in span. Then there exist $s, \sigma \in \mathbb{Z}_+$ such that $g(s, \sigma+1) \leq g(s, \sigma)$. Fix an integer $q \geq 3$. For each large integer m, construct a hierarchy h_m with $|V(h_m)| = qm$ containing exactly $K_m = m$ *targets* each with depth s and span σ, as follows.

Construct s shared ancestors (without loss, s distinct roots, each with an edge to every target) and a shared collection of σ leaves. If $\sigma = 0$, omit these leaves. Each target points to all σ shared leaves. If $\sigma \geq 1$, choose a distinguished leaf b_1. Arrange the remaining wiring so that the set of nodes that can reach b_1 is exactly the set of targets, the s shared ancestors, and a support set R_m satisfying $|R_m| \leq \bar{c}$ for some fixed $\bar{c}$ independent of m.[2] If $\sigma = 0$, set $R_m = \varnothing$ (Figure 12). Finally, designate at least one remaining node of h_m to be isolated (having no incoming or outgoing edges); this is possible for all sufficiently large m because

[2]The proof still holds if we set $R_m = \varnothing$ (using only direct edges to form the required depth and span). We include bounded support sets in the formal construction to illustrate that the limit infimum bounding argument is robust to arbitrary internal bridging architectures, provided the number of such bridging nodes remains independent of m.

the number of remaining nodes is $qm - m = (q-1)m$, which grows to infinity, exceeding the number of required non-target nodes.

Now form a compound $h'_m \in h_m \uplus h_m$, where $\uplus$ is the compounding operator in IPG. Choose $z^{\downarrow}$ to be an isolated node in the lower copy. If $\sigma \geq 1$, let $b_1^{\uparrow}$ be the copy of b_1 in the upper copy and add the single cross-edge $b_1^{\uparrow} \to z^{\downarrow}$. If $\sigma = 0$, add cross-edges from each upper-copy target to $z^{\downarrow}$. In either case, every upper-copy target gains $z^{\downarrow}$ as a new descendant, so its span increases by exactly 1 and its depth is unchanged.

Because $z^{\downarrow}$ has no outgoing edges, the only new reachability relations are paths ending at $z^{\downarrow}$. Hence, a node's span increases if and only if it could reach the linkage tail(s) before the link: $b_1^{\uparrow}$ when $\sigma \geq 1$, and the targets themselves when $\sigma = 0$. By construction, the nodes with $\Delta\sigma > 0$ are exactly the targets, the s shared ancestors, the support set R_m, and the linkage tail $b_1^{\uparrow}$ when $\sigma \geq 1$. Define the residual set

$$\widetilde{R}_m = (\text{both copies of } R_m) \cup \{z^{\downarrow}\} \cup (\{b_1^{\uparrow}\} \text{ if } \sigma \geq 1) \cup (\text{the } s \text{ shared ancestors in the upper copy}).$$

Then

$$|\widetilde{R}_m| \leq s + 2\bar{c} + 2 = c_0,$$

and c_0 is a finite constant independent of m. Moreover, every node outside the targets and $\widetilde{R}_m$ has $\Delta s = \Delta\sigma = 0$ and hence $\Delta g_m(v) = 0$.

Since $\frac{K_m}{|V(h'_m)|} = \frac{m}{2qm} = \frac{1}{2q} > 0$, the ratio of targets to total nodes is exactly constant for all m. Hence the sequence is macroscopic (with macroscopic constant $1/(2q)$).

For each target v we have $\Delta g_m(v) = g(s, \sigma+1) - g(s, \sigma) \leq 0$. Thus every node with $\Delta g_m(v) > 0$ lies in $\widetilde{R}_m$, so there are at most c_0 nodes with strictly positive gains. Let $\Delta g_m^+(v) := \max\{\Delta g_m(v), 0\}$ and let $\Delta g_m^+(1) \geq \Delta g_m^+(2) \geq \cdots$ be the order statistics. Because at most c_0 of the $\Delta g_m^+(v)$ are nonzero, the total positive gain equals $\sum_{i=1}^{c_0} \Delta g_m^+(i)$, and by definition of $M_m(c)$ in IPG we have

$$\frac{1}{|V(h'_m)|} \sum_{v \in V(h'_m)} \Delta g_m^+(v) = \frac{1}{|V(h'_m)|} \sum_{i=1}^{c_0} \Delta g_m^+(i) \leq M_m(c_0).$$

Therefore,

$$I(h'_m) - I(h_m) - M_m(c_0) = \frac{1}{|V(h'_m)|} \sum_{v \in V(h'_m)} \Delta g_m(v) - M_m(c_0) \leq \frac{1}{|V(h'_m)|} \sum_{v \notin \widetilde{R}_m} \Delta g_m(v) \leq 0,$$

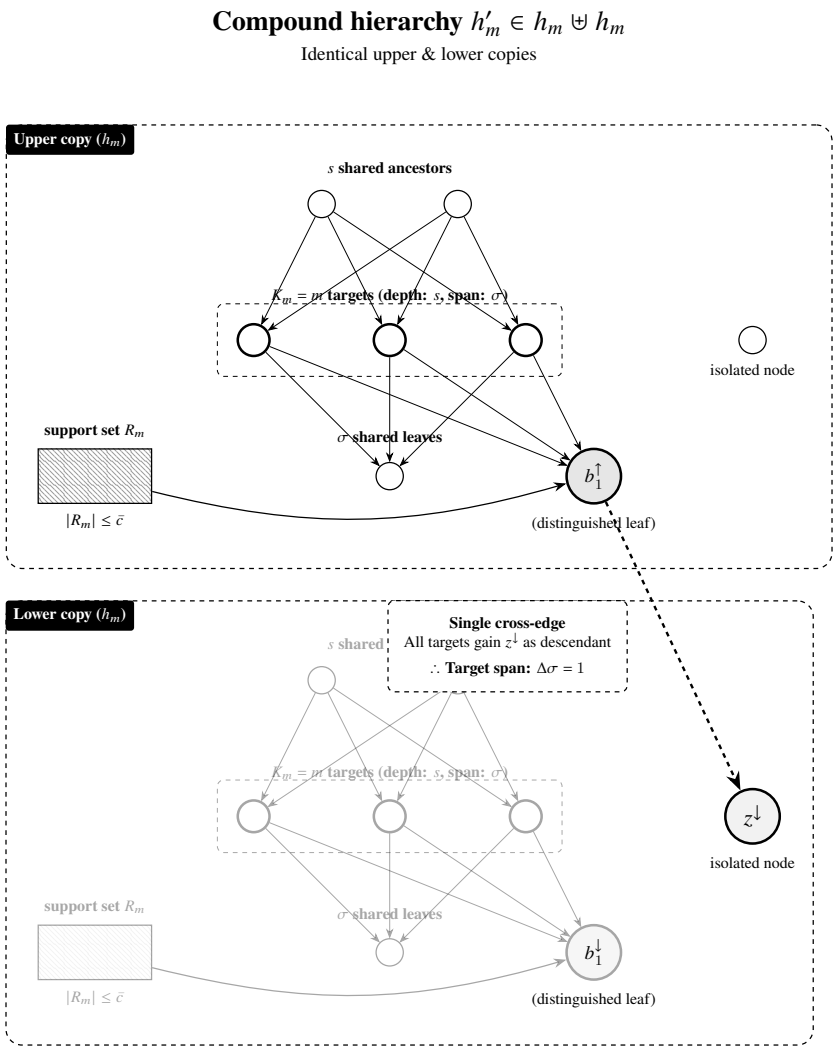

Figure 12: Compound hierarchy h'_m.

since nodes outside $\widetilde{R}_m$ are either targets (nonpositive gains) or unchanged nodes (zero gain). Because the sequence is macroscopic, taking $\liminf_{m\to\infty}$ strictly contradicts IPG. Hence $g(s,\sigma+1) > g(s,\sigma)$ for all s,σ.

Step 2 (strict increase in depth). A symmetric construction yields strict increase in depth. If for some s,σ we had $g(s+1,\sigma) \le g(s,\sigma)$, build h_m with $K_m = m$ targets at depth s and span σ so that, for $s \ge 1$, there is a root a_1 whose set of descendants is exactly the K_m targets, the σ shared leaves, and a support set R'_m with $|R'_m| \le \bar{c}$ (and all other nodes lie outside $\mathrm{Desc}(a_1)$) (Figure 13). If $s = 0$, omit a_1 and take $R'_m = \varnothing$.

Form $h'_m \in h_m \uplus h_m$ and add a cross-edge from an isolated node $w^{\uparrow}$ in the upper copy to $a_1^{\downarrow}$ in the lower copy (or to each target if $s = 0$). Since $w^{\uparrow}$ has no incoming edges, the only new reachability relations are paths originating at $w^{\uparrow}$, so depth increases for the targets, the σ shared leaves, and R'_m. Node $a_1^{\downarrow}$ also gains $w^{\uparrow}$ as a new ancestor. Define

$$\widetilde{R}'_m = (\text{both copies of } R'_m) \cup \{w^{\uparrow}\} \cup (\{a_1^{\downarrow}\} \text{ if } s \ge 1) \cup (\text{the } \sigma \text{ shared leaves in the lower copy}).$$

Then $|\widetilde{R}'_m| \le \sigma + 2\bar{c} + 2 = c'_0$, a finite constant independent of m, and every node with $\Delta g_m(v) > 0$ lies in $\widetilde{R}'_m$.

Applying the Step-1 order-statistic argument with residual size c'_0 yields

$$I(h'_m) - I(h_m) - M_m(c'_0) \le 0.$$

Just as in Step 1, taking $\liminf_{m\to\infty}$ directly contradicts IPG along the macroscopic sequence. Therefore $g(s+1,\sigma) > g(s,\sigma)$ for all s,σ.

Step 3 (marginals bounded away from zero along fixed coordinates). Fix s. Suppose $\inf_\sigma \Delta_\sigma g(s,\sigma) = 0$, where $\Delta_\sigma g(s,\sigma) = g(s,\sigma+1) - g(s,\sigma)$. Then there exists a sequence $\sigma_k \to \infty$ such that $\Delta_\sigma g(s,\sigma_k) \to 0$. Using the span gadget from Step 1 with targets at fixed depth s and span $\sigma_m = \sigma_{k(m)}$, choose $k(m)$ increasing such that

$$s + \sigma_{k(m)} + \bar{c} + 1 \le (q-1)m,$$

ensuring the construction fits within size qm while reserving an isolated node in h_m.

The same residual set bound gives a fixed constant c_0 (depending only on s and $\bar{c}$) such

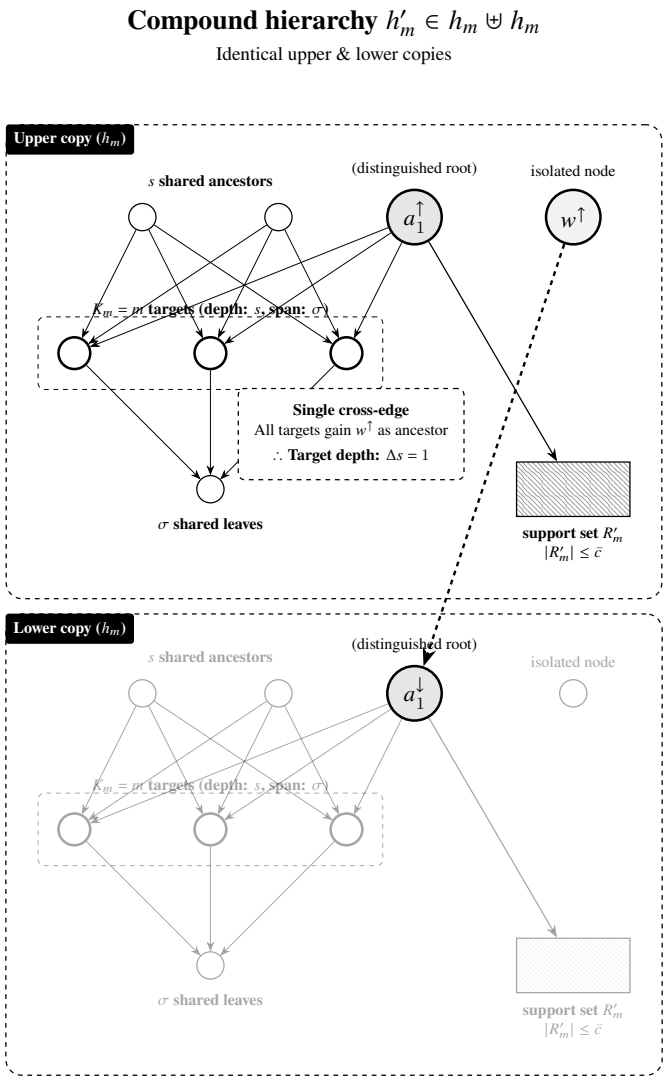

Figure 13: Compound hierarchy h'_m.

that all strictly positive gains lie among at most c_0 nodes. Hence

$$I(h'_m) - I(h_m) - M_m(c_0) \leq \frac{K_m}{|V(h'_m)|} \Delta_\sigma g(s, \sigma_m).$$

Since $K_m/|V(h'_m)| = 1/(2q)$, and $\Delta_\sigma g(s, \sigma_m) \to 0$, the right-hand side converges to 0, so the $\liminf$ is ≤ 0, contradicting IPG. Therefore $\inf_\sigma \Delta_\sigma g(s, \sigma) > 0$.

Fix σ and argue symmetrically using the depth gadget from Step 2: if $\inf_s \Delta_s g(s, \sigma) = 0$ with $\Delta_s g(s, \sigma) = g(s+1, \sigma) - g(s, \sigma)$, choose a sequence $s_k \to \infty$ with $\Delta_s g(s_k, \sigma) \to 0$, and then choose an increasing map $k(m)$ such that

$$s_{k(m)} + \sigma + \bar{c} + 1 \leq (q-1)m.$$

Placing the targets at depth $s_{k(m)}$ and span σ and applying the same reasoning (now with a fixed residual bound depending only on σ and $\bar{c}$) yields $\inf_s \Delta_s g(s, \sigma) > 0$ for each fixed σ.

This completes the proof. ■

4.3.2 Axiomatic characterization

With the foundational structure and regularity conditions established, we can now state the main representation theorem for the family of average bilateral indices.

Theorem 3. *A hierarchical index $I : \mathscr{H} \to \mathbb{R}$ satisfies A, AS, and ETN, and its corresponding bilateral average representation satisfies IPG, if and only if there exists a function $g : \mathbb{Z}_+ \times \mathbb{Z}_+ \to \mathbb{R}$ such that for all $h \in \mathscr{H}$,*

$$I(h) = \frac{1}{|V(h)|} \sum_{v \in V(h)} g\big(s_h(v), \sigma_h(v)\big),$$

and g has discrete marginals strictly bounded away from zero: there exists a constant $m > 0$ such that for all $s, \sigma \in \mathbb{Z}_+$,

$$m \leq \Delta_s g(s, \sigma) \qquad \text{and} \qquad m \leq \Delta_\sigma g(s, \sigma).$$

Remark 8. Although not explicitly listed as a separate regularity condition in the theorem statement, the requirement that the discrete marginals of g are bounded below by a strictly positive constant $m > 0$ necessarily implies that the local evaluation function g is strictly increasing in both its depth (s) and span (σ) arguments. Because every unit coordinate shift

guarantees a strict evaluation gain of at least m, g must strictly monotonically increase along both dimensions.

Proof of Theorem 3. **Necessity (⇒).** Suppose I satisfies A, AS, ETN, and its corresponding bilateral average representation satisfies IPG. By Lemma 1, the combination of A, AS, and ETN implies that I is a bilateral average index, admitting a representation of the form

$$I(h) = \frac{1}{|V(h)|} \sum_{v \in V(h)} g(s_h(v), \sigma_h(v))$$

for some real-valued local evaluation function g.

Because I satisfies IPG, Proposition 1 dictates that g must be strictly increasing. We now show IPG implies the discrete marginals are uniformly bounded away from zero. Suppose, toward a contradiction, that $\inf_{s,\sigma} \Delta_s g(s, \sigma) = 0$. Then there exists a sequence of coordinate pairs (s_k, σ_k) such that $\Delta_s g(s_k, \sigma_k) \to 0$ as $k \to \infty$.

Following the gadget construction in the proof of Proposition 1, we can build a macroscopic sequence of compounds $h'_k \in h_k \uplus h_k$, where $\uplus$ is the partial-append operator (adding leaf-to-root cross-edges), applied here to two copies of h_k (one designated as the "upper" copy, one as the "lower" copy). Choose the gadget parameters such that the number of target nodes K_k satisfies $\lim_{k\to\infty} K_k/|V(h'_k)| = \bar{\rho}$ for some constant $\bar{\rho} > 0$. For each k, the hierarchy contains exactly K_k target nodes placed at depth s_k and span σ_k. Furthermore, the construction guarantees that all strictly positive coordinate shifts outside the targets are confined to a residual set $\widetilde{R}_k$ whose size is strictly bounded by a finite constant c_0 that is completely independent of k.

Identify each node $v \in V(h'_k)$ with its counterpart v_0 in the appropriate baseline copy of h_k. Since I is an AS-average, replication invariance yields $I(h_k) = I(h_k \cup h_k)$ (where $\cup$ denotes the disjoint union of the two baseline copies). Note that $|V(h'_k)| = |V(h_k \cup h_k)| = 2|V(h_k)|$. Hence, the difference in the index can be expressed precisely as the average of node-level evaluation increments:

$$I(h'_k) - I(h_k) = \frac{1}{|V(h'_k)|} \sum_{v \in V(h'_k)} \Delta g_k(v).$$

Let T_k be the set of targets. By construction, every node v outside $T_k \cup \widetilde{R}_k$ has

$\Delta s_k(v) = \Delta\sigma_k(v) = 0$, and hence $\Delta g_k(v) = 0$. Thus,

$$I(h'_k) - I(h_k) = \frac{1}{|V(h'_k)|}\left(\sum_{v\in T_k} \Delta g_k(v) + \sum_{v\in \widetilde{R}_k} \Delta g_k(v)\right).$$

Moreover, $\Delta g_k(v) \leq \Delta g_k^+(v) = \max\{\Delta g_k(v), 0\}$ for all v, which implies:

$$I(h'_k) - I(h_k) \leq \frac{1}{|V(h'_k)|}\left(\sum_{v\in T_k} \Delta g_k(v) + \sum_{v\in \widetilde{R}_k} \Delta g_k^+(v)\right).$$

Because $|\widetilde{R}_k| \leq c_0$, the sum of the positive gains in $\widetilde{R}_k$ is bounded by the sum of the c_0 largest positive gains across the entire hierarchy. Let

$$\Delta g^+_{k,(1)} \geq \Delta g^+_{k,(2)} \geq \cdots$$

denote the order statistics of the positive gains in h'_k. Then

$$\sum_{v\in \widetilde{R}_k} \Delta g_k^+(v) \leq \sum_{i=1}^{c_0} \Delta g^+_{k,(i)}.$$

Dividing by $|V(h'_k)|$, this normalized sum is bounded directly by the definition of the outlier term $M_k(c_0)$ in IPG. Hence, we obtain the upper bound:

$$I(h'_k) - I(h_k) - M_k(c_0) \leq \frac{1}{|V(h'_k)|}\sum_{v\in T_k} \Delta g_k(v).$$

Because every target node experiences a unit increase in depth and no change in span, its individual evaluation gain is exactly $\Delta_s g(s_k, \sigma_k) > 0$ (since g was established as strictly increasing). Therefore,

$$I(h'_k) - I(h_k) - M_k(c_0) \leq \frac{K_k}{|V(h'_k)|}\Delta_s g(s_k, \sigma_k).$$

Because the sequence is macroscopic (with $\lim_{k\to\infty} K_k/|V(h'_k)| = \bar{\rho} > 0$) and we assumed the marginal gains shrink to zero ($\Delta_s g(s_k, \sigma_k) \to 0$), evaluating the intensive growth

condition yields:

$$\liminf_{k\to\infty}\left[I(h'_k) - I(h_k) - M_k(c_0)\right] \le \lim_{k\to\infty} \frac{K_k}{|V(h'_k)|}\Delta_s g(s_k, \sigma_k) = 0.$$

This implies the macroscopic sequence fails to generate strictly positive intensive growth, directly contradicting IPG. Hence, $\inf_{s,\sigma} \Delta_s g(s,\sigma) > 0$. An identical argument applying the span gadget yields $\inf_{s,\sigma} \Delta_\sigma g(s,\sigma) > 0$. Letting

$$m = \min\left\{\inf_{s,\sigma} \Delta_s g(s,\sigma), \inf_{s,\sigma} \Delta_\sigma g(s,\sigma)\right\} > 0$$

proves the existence of a uniform strictly positive lower bound m across the entire domain, establishing necessity.

Sufficiency ($\Leftarrow$). Suppose I admits the displayed representation where g has discrete marginals bounded below by a strictly positive constant $m > 0$. By the sufficiency direction of Lemma 1, any index possessing this functional dependence strictly on depth and span coordinates necessarily satisfies A, AS, and ETN.

To establish IPG, note first that because each compound $h'_k \in h_k \uplus h_k$ is formed by adding directed edges and deleting none, every node's depth and span are weakly nondecreasing (since adding edges preserves all existing directed paths, so ancestor and descendant sets can only expand): $\Delta s_k(v) \ge 0$ and $\Delta\sigma_k(v) \ge 0$ for all v. The total evaluation gain for any node can be decomposed into a telescoping sequence of unit steps (with the convention that an empty sum equals 0):

$$g(s+\Delta s, \sigma+\Delta\sigma) - g(s,\sigma) = \sum_{i=0}^{\Delta s-1} \Delta_s g(s+i,\sigma) + \sum_{j=0}^{\Delta\sigma-1} \Delta_\sigma g(s+\Delta s, \sigma+j).$$

Consequently, the uniform lower bound $m > 0$ implies that any strictly positive coordinate shift ($\Delta s + \Delta\sigma \ge 1$) yields a marginal evaluation gain of at least m. For any macroscopic sequence of compounds ($h'_k \in h_k \uplus h_k$), a non-vanishing fraction $\rho' > 0$ of nodes experiences such a shift. Thus, there are at least $\rho'|V(h'_k)|$ nodes that achieve a positive evaluation gain of at least m.

The outlier term $M_k(c)$ explicitly removes the c largest marginal gains from the normalized aggregate sum. Even if all c removed nodes are drawn directly from this macroscopic set, there remain at least $\rho'|V(h'_k)| - c$ nodes contributing a gain of at least m to the residual

sum. Therefore, evaluating the full intensive growth condition yields:

$$\liminf_{k\to\infty}\left[I(h'_k) - I(h_k) - M_k(c)\right] \geq \lim_{k\to\infty}\frac{\rho'|V(h'_k)| - c}{|V(h'_k)|}m = \rho' m > 0,$$

so the index successfully leaves a persistent per-capita structural trace and satisfies IPG. This establishes sufficiency. ■

Thus, the bilateral average framework provides a precise local evaluation that strictly responds to macroscopic growth. By acting as a coordinatewise increasing average, the index precludes asymptotic saturation (vanishing marginals) and remains robust to finite localized noise.

Remark 9. Admissible specifications and the role of IPG. Theorem 3 imposes a *one-sided* restriction on $g(s,\sigma)$: along *any* macroscopic sequence of compounds, the per-capita gain must remain bounded away from zero after discounting any finite set of maximal node-level gains via $M_k(c)$. By preventing macroscopic growth from either decaying to zero or being driven entirely by a finite subset of nodes, IPG rules out two broad classes of functional behavior:

- *Saturation (bounded functions):* If g is strictly increasing but bounded (e.g., $g(s,\sigma) = 1 - e^{-(s+\sigma)}$), its discrete marginals must vanish as $s,\sigma \to \infty$. Because IPG holds for *all* macroscopic sequences, we can construct a sequence where structural additions are localized at arbitrarily large coordinates. Along such a sequence, the gross per-capita gain $I(h'_k) - I(h_k)$ decays to zero. Since $M_k(c) \geq 0$, the adjusted intensive growth $\liminf_{k\to\infty}\left[I(h'_k) - I(h_k) - M_k(c)\right]$ cannot exceed zero, directly contradicting IPG.
- *Sublinear unbounded growth:* If g is unbounded but exhibits vanishing marginals (e.g., $g(s,\sigma) = \sqrt{s}+\sqrt{\sigma}$), its discrete gains still decay to zero as $s,\sigma \to \infty$. By constructing a macroscopic sequence where a fixed fraction of nodes experiences coordinate shifts at arbitrarily large initial coordinates, the gross per-capita gain $I(h'_k) - I(h_k)$ is driven to zero. Since $M_k(c) \geq 0$, this guarantees $\liminf_{k\to\infty}\left[I(h'_k) - I(h_k) - M_k(c)\right] \leq 0$, which contradicts IPG.

Why IPG permits explosive curvature. Superlinear specifications, such as $g(s,\sigma) = s^2 + \sigma^2$, satisfy IPG because the condition does not restrict the maximum size of marginal gains. It only requires that a strictly positive per-capita increase remains after discarding any finite number c of the largest gains. Under a convex g, a macroscopic fraction of nodes experiences strictly increasing marginal gains as coordinates grow. Removing c outliers

leaves an overwhelming majority of these nodes intact, ensuring the trimmed per-capita gain remains strictly positive. Consequently, IPG acts as a lower bound on systemic growth rather than a restriction on extreme upper-tail behavior.

Finally, note that while applying any strictly increasing transformation $T(\cdot)$ to the aggregate index—yielding $J(h) = T(I(h))$—preserves the induced ordinal ranking and properties like SR, it generally breaks AS. Indeed, AS requires an arithmetic average of localized evaluations, and only positive affine transformations ($J = aI + b$) preserve this structure. Thus, the additive representation in Theorem 3 is unique up to a positive affine transformation.

Remark 10. Equivalence of unilateral and bilateral specifications. Superficially, Theorem 3 appears to rule out "unilateral" indices depending solely on depth ($g(s, \sigma) = f(s)$) or span ($g(s, \sigma) = f(\sigma)$), as they are not strictly increasing in the omitted argument. However, because every ancestor-descendant pair contributes equally to total depth and total span, any hierarchy satisfies the identity

$$\sum_{v \in V(h)} s_h(v) = \sum_{v \in V(h)} \sigma_h(v).$$

Consequently, any unilateral specification

$$I(h) = \frac{1}{|V(h)|} \sum_{v \in V(h)} f(s_h(v))$$

can be mapped to an equivalent bilateral specification via a zero-sum shift:

$$\hat{g}(s, \sigma) = \beta f(s) + \gamma + \alpha(\sigma - s), \quad \beta > 0.$$

The resulting index, $\hat{I}(h) = \beta I(h) + \gamma$, preserves the ordinal ranking of hierarchies. Furthermore, because $s_h(v)$ and $\sigma_h(v)$ depend exclusively on the localized environment $N(v, h)$, this transformation safely preserves AS.

For $\hat{g}$ to satisfy the strict monotonicity required by Theorem 3, its discrete marginals $\Delta_\sigma \hat{g}(s, \sigma) = \alpha$ and $\Delta_s \hat{g}(s, \sigma) = \beta[f(s+1) - f(s)] - \alpha$ must be strictly positive. Such constants $\alpha, \beta > 0$ exist if and only if $\inf_{s \geq 0}[f(s+1) - f(s)] > 0$. If f exhibits vanishing marginals (e.g., $f(s) = \sqrt{s}$), the term $\beta[f(s+1) - f(s)]$ eventually falls below α for large s, rendering $\hat{g}$ decreasing in depth. This algebraic constraint perfectly mirrors IPG: any unilateral function satisfying IPG (e.g., $f(s) = s$) possesses marginals bounded away from

zero. Therefore, it can always be recast as a strictly increasing bilateral evaluation (e.g., setting $\beta = 1, \alpha = 0.5$ transforms $f(s) = s$ into $\hat{g}(s, \sigma) = 0.5s + 0.5\sigma$). Thus, the bilateral framework subsumes admissible unilateral indices.

4.3.3 Non-redundancy and scope of the assumptions

The assumptions in Theorem 3 are nested by design. AS introduces a witness f on environments, and ETN is formulated as a restriction on that witness. Moreover, IPG is defined strictly for bilateral average indices (Definition 2). Lemma 1 provides the canonical reduction to this class via A+AS+ETN.

Because of this typing, strict "logical independence" of all four statements is not well-posed under the baseline definitions: dropping AS removes the domain on which ETN is evaluated, and dropping ETN removes the canonical reduction (Lemma 1) that yields the universal $g(s, \sigma)$-representation used to place indices in the bilateral-average class. Instead, we establish non-redundancy by providing, for each assumption, an index that satisfies the remaining well-scoped assumptions and exhibits the pathology that the dropped assumption is meant to exclude. Throughout, a "macroscopic sequence of compounds" means that a non-vanishing fraction of nodes experiences a positive coordinate shift.

Dropping A. Assign weights $w(v) = 2$ if the label of v is even and $w(v) = 1$ otherwise, and define the weighted index

$$I_{\mathrm{w}}(h) = \frac{1}{|V(h)|} \sum_{v \in V(h)} w(v)\big(s_h(v) + \sigma_h(v)\big).$$

This index satisfies AS with witness $f(N(v, h)) = w(v)(s_h(v) + \sigma_h(v))$. It violates A because relabeling that swaps an even-labeled node with an odd-labeled one changes the weights and thus the index.

This index also satisfies ETN: any ETN-admissible rewiring of $N(v, h)$ preserves the ancestor and descendant sets (denoted $\mathrm{Anc}_h(v)$ and $\mathrm{Desc}_h(v)$) and hence their cardinalities $s_h(v)$ and $\sigma_h(v)$; since $w(v)$ depends only on v's label (which is unchanged), $f(N(v, h)) = w(v)\big(s_h(v) + \sigma_h(v)\big)$ remains invariant.

Since IPG is defined only for bilateral average indices (Definition 2), i.e., indices admitting a graph-independent representation $I(h) = \frac{1}{|V(h)|} \sum_v g(s_h(v), \sigma_h(v))$ (which automatically implies A), it is not formally evaluated for this label-dependent counterexample.

Dropping AS. Consider the geometric-mean index

$$I_{\mathrm{gm}}(h) = \prod_{v \in V(h)} \big(1 + s_h(v) + \sigma_h(v)\big)^{\frac{1}{|V(h)|}}.$$

This index satisfies A and depends only on the multiset of pairs $(s_h(v), \sigma_h(v))$. In particular, it is insensitive to within-environment routing whenever ancestor and descendant *sets* are preserved (capturing the intuition that only absolute ancestor/descendant sets matter, not internal routing). However, it violates AS.

Let h_1 be the single-node hierarchy and let h_2 be the two-node chain. Then $I_{\mathrm{gm}}(h_1) = 1$ and $I_{\mathrm{gm}}(h_2) = 2$ (each node in the chain has $s + \sigma = 1$). For the disjoint union $h = h_1 \cup h_2$,

$$I_{\mathrm{gm}}(h) = \big(1 \cdot 2 \cdot 2\big)^{1/3} = 2^{2/3}.$$

If I_{gm} satisfied AS with a single global witness f, then in the disjoint union the environments of nodes in h_1 (resp. h_2) would be identical to their environments in their original components (since no cross-component paths exist). This implies the weighted arithmetic-mean decomposition

$$I_{\mathrm{gm}}(h_1 \cup h_2) = \frac{|V(h_1)|}{|V(h_1)| + |V(h_2)|} I_{\mathrm{gm}}(h_1) + \frac{|V(h_2)|}{|V(h_1)| + |V(h_2)|} I_{\mathrm{gm}}(h_2) = \frac{1}{3} \cdot 1 + \frac{2}{3} \cdot 2 = \frac{5}{3},$$

which contradicts $2^{2/3} \neq 5/3$.

Dropping ETN. Let $m_h(v)$ be the number of edges whose endpoints both lie in the environment $N(v, h)$ (i.e., edges strictly internal to the induced subgraph on that environment). Define the routing-sensitive index

$$I_{\mathrm{rt}}(h) = \frac{1}{|V(h)|} \sum_{v \in V(h)} \Big(s_h(v) + \sigma_h(v) + m_h(v)\Big).$$

This index satisfies A and AS (with witness $f(N(v, h)) = s_h(v) + \sigma_h(v) + m_h(v)$). It violates ETN because $m_h(v)$ can change under rewiring that preserves v's ancestor and descendant *sets*. For example, if v has two incomparable root ancestors a_1 and a_2 (no directed path between them), adding the edge $a_1 \to a_2$ preserves $\mathrm{Anc}_h(v)$ and $\mathrm{Desc}_h(v)$ and keeps the environment acyclic, but increases $m_h(v)$ by 1 while leaving $s_h(v)$ and $\sigma_h(v)$ unchanged.

Since IPG is defined only for bilateral average indices (Definition 2), and Lemma 1

is the reduction that yields such a g-representation under A+AS+ETN, it is not formally evaluated for this routing-sensitive example without additional reduction structure. The example isolates the distinct normative content of ETN: the local evaluation should depend only on who the ancestors and descendants of the center node are, not on how those ancestors and descendants are connected among themselves.

Dropping IPG. Consider the saturating bilateral average index

$$I_{\mathrm{sat}}(h) = \frac{1}{|V(h)|} \sum_{v \in V(h)} g\big(s_h(v), \sigma_h(v)\big), \quad g(s,\sigma) = 2 - (s+1)^{-1} - (\sigma+1)^{-1}.$$

This index satisfies A, AS, and ETN (it depends only on (s,σ)). However, its unit marginals

$$\Delta_s g(s,\sigma) = (s+1)^{-1}(s+2)^{-1}, \quad \Delta_\sigma g(s,\sigma) = (\sigma+1)^{-1}(\sigma+2)^{-1},$$

vanish as coordinates grow. Fix any $\varepsilon > 0$. Using the standard target-and-trunk constructions from previous proofs, choose baseline hierarchies h_k to place a macroscopic fraction $\rho > 0$ of nodes (in h'_k, equivalently in each copy of h_k) at coordinates (s,σ) sufficiently large that every unit marginal of g is at most ε. We then form compounds $h'_k \in h_k \uplus h_k$ such that exactly one coordinate of these targeted nodes increases by one. The construction is chosen so that any spillover shifts are confined to a residual set R_k of uniformly bounded size: there exists a constant $c_0 \geq 1$ such that $|R_k| \leq c_0$ for all k (as in the target-and-trunk gadgets used earlier). Evaluating IPG at the fixed integer $c = c_0$, the trimming term $M_k(c_0)$ removes the c_0 largest positive gains. Since each targeted node has gain $\Delta g_k(v) \leq \varepsilon$, any spillover node with gain $> \varepsilon$ must be among these top c_0 gains and is therefore trimmed. Any remaining spillover gains are $\leq \varepsilon$ and hence contribute at most $c_0\varepsilon/|V(h'_k)| \to 0$ (since $|V(h'_k)| \to \infty$) to the per-capita difference. Hence, after trimming we have

$$I_{\mathrm{sat}}(h'_k) - I_{\mathrm{sat}}(h_k) - M_k(c_0) \leq \rho\,\varepsilon + \frac{c_0\varepsilon}{|V(h'_k)|}.$$

Taking $\liminf_{k\to\infty}$ yields $\leq \rho\,\varepsilon$, and since $\varepsilon > 0$ is arbitrary, the limit inferior cannot be strictly positive. Thus IPG fails; hence it is a non-redundant growth restriction that rules out asymptotically saturating local evaluations even when A, AS, and ETN hold.

4.3.4 Compliance with the axiomatic framework

To complete the analysis of the average bilateral indices characterized in Theorem 3, we summarize their compliance with the broader set of axioms considered in this paper.

Core characterization. By Theorem 3, every index in this family takes the form

$$I(h) \;=\; \frac{1}{|V(h)|} \sum_{v \in V(h)} g\big(s_h(v), \sigma_h(v)\big),$$

where g is strictly increasing in each argument (indeed, its discrete marginals are uniformly bounded away from zero). Consequently, the family satisfies A, AS, ETN, and IPG.

Subordination Removal. The strict coordinatewise monotonicity of g guarantees SR. Removing a subordination relation cannot create new reachability. Indeed, any shortcut edge (p, i) added by Definition 1 links a node p that was already an ancestor of i in h (via $p \to j \to i$), so it cannot create a new reachable pair; it can only preserve reachability that would otherwise be lost when (j, i) is deleted. Thus, every node's depth and span are weakly non-increasing. If the removed edge is redundant, the reachability relation is unchanged, so no ancestor/descendant sets change and hence $I(h) = I(h')$, satisfying the weak part of SR. If the removed edge is non-redundant, at least one node's depth or span strictly decreases (in particular, the removed supervisor is removed from the subordinate's ancestor set), and since $|V(h)|$ is unchanged and g is strictly increasing, the index strictly decreases, as required by SR.

Relaxed extension axioms. This family satisfies the partial extension axioms UE* and DE*. For any hierarchies h_o, h and any extension $h^* \in h_o \uplus h$, the construction of $\uplus$ adds at least one leaf-to-root cross-edge. Consequently, at least one node's depth or span strictly increases, and none decrease (as adding edges only expands ancestor/descendant sets). Since g is strictly increasing coordinatewise, the numerator $\sum_{v \in V(h^*)} g\big(s_{h^*}(v), \sigma_{h^*}(v)\big)$ is strictly greater than the corresponding sum in the disjoint union $h_o \cup h$. The index of the disjoint union $h_o \cup h$ is exactly the population-weighted average of $I(h_o)$ and $I(h)$. If $I(h_o) \geq I(h)$ (the weak dominance premise of UE*/DE*), then $I(h_o \cup h) \geq I(h)$. Because the added cross-edges in h^* strictly increase the evaluation sum while maintaining the exact same total population size (i.e., $|V(h^*)| = |V(h_o \cup h)| = |V(h_o)| + |V(h)|$), it follows that $I(h^*) > I(h_o \cup h) \geq I(h)$, thereby satisfying both axioms.

Table 1: Characterization and axiomatic compliance of the average bilateral index family

Family characterization (Theorem 3)		**Description**
Functional form		$I(h) = \frac{1}{\lvert V(h)\rvert} \sum_{v \in V(h)} g\big(s_h(v), \sigma_h(v)\big)$
Properties of g		The local evaluation function g depends on a node's depth (s) and span (σ). It is strictly increasing in both dimensions, meaning any unit increase in depth or span yields a strict evaluation gain. Specifically, its discrete marginals are uniformly bounded below by a strictly positive constant ($m > 0$).
Axiomatic Compliance	**Status**	**Reason**
A (Anonymity)	✓	Evaluation depends only on structural depth and span, invariant under relabeling.
AS (Additive Separability)	✓	Index is defined as a population average of localized node evaluations.
ETN (Env. Topology Neutrality)	✓	Local score determined solely by ancestor/descendant counts, independent of internal routing.
IPG (Intensive Pop. Growth)	✓	Discrete marginals of g are uniformly bounded away from zero.
SR (Subordination Removal)	✓	Strict coordinatewise monotonicity ensures index weakly/strictly decreases upon edge removal.
UE* / DE* (Relaxed Extensions)	✓	Weak dominance preconditions isolate hierarchical gains from population dilution.
UE / DE (Vertical Extensions)	×†	For regular indices, appending large flat structures dilutes the average.
QS (Quadratic Separability)	×	Incompatible with intensive averaging; density normalization scales inversely with replication.
ADD (Additivity)	×	Incompatible with intensive averaging; index sum scales with replication.

†Under the regularity hypotheses of Theorem 1 (in particular, bounded marginal returns).

Unrestricted extension axioms. By the impossibility result in Theorem 1, no regular AS index—and hence no regular member of the present average-bilateral family—can satisfy unrestricted vertical extensions UE and DE. For any index with bounded marginal returns, appending a sufficiently large flat structure dilutes the population average, eventually forcing a decreased hierarchical degree for some admissible extension, violating both UE and DE.

Quadratic Separability and concentration. The axiom QS evaluates hierarchy as a global density via a quadratic normalization, which is antagonistic to the present intensive family. Because average bilateral indices are replication-invariant ($I(h^k) = I(h)$), they cannot satisfy QS, which requires the index of a replicated hierarchy to scale inversely with the replication factor k (for non-flat cases where the index evaluates to a strictly positive numerator).

Additivity. Finally, the family violates ADD (Additivity). Additivity requires a strict extensive sum of localized evaluations,

$$I(h) = \sum_{v \in V(h)} f\big(N(v, h)\big),$$

whereas the present family is an average normalized by $|V(h)|$. Accordingly, replication does not scale the value: if h^k is the disjoint union of k copies of h, then $I(h^k) = I(h)$, while ADD would require $I(h^k) = kI(h)$.

4.4 A class of total bilateral indices

While Theorem 3 establishes a family of *average* bilateral indices, certain analytical contexts require measuring the *aggregate* structural volume of an organization. Moving from average to total indices requires replacing the intensive, population-averaged framework of AS with its extensive counterpart, ADD.

As in the intensive case, we impose ETN to exclude sensitivity to internal routing among a node's supervisors and subordinates. Here ETN is evaluated relative to the ADD witness f.

Lemma 2. *A hierarchical index $I : \mathscr{H} \to \mathbb{R}$ satisfies* A, ADD, *and* ETN *if and only if there*

exists a function $g : \mathbb{Z}_+ \times \mathbb{Z}_+ \to \mathbb{R}$ *such that for every* $h \in \mathscr{H}$,

$$I(h) = \sum_{v \in V(h)} g\big(s_h(v), \sigma_h(v)\big).$$

The argument is analogous to the proof of Lemma 1, with AS replaced by ADD. Because ADD aggregates via a node-level sum rather than a population average, the combinatorial logic is unaffected by the absence of the $1/|V(h)|$ global scalar. In particular, the same three steps apply: (i) symmetrize the ADD witness f using A to obtain an isomorphism-invariant $\bar{f}$; (ii) use ETN to purely locally rewire each environment to the canonical double-star $D_{s,\sigma}$ without changing $\bar{f}$; and (iii) define $g(s, \sigma) = \bar{f}(D_{s,\sigma})$ to obtain the displayed representation.

By Lemma 2, the axioms A, ADD, and ETN jointly imply bilateral topological invariance in the extensive setting. Because the argument proceeds node by node without a population denominator, the index must take the form of an unweighted sum.

In the intensive framework, an arbitrary baseline evaluation for an isolated node merely shifts the overall population average by a constant, leaving relative structural comparisons intact. However, when measuring aggregate hierarchy, it becomes natural to impose a zero-baseline. If an isolated individual with no superiors or subordinates carried a non-zero structural evaluation, the total hierarchy of an organization would spuriously scale with the raw size of its unstructured workforce. To prevent this, we formally require that isolated nodes contribute no structural volume.

Isolated Node Normalization (INN). A hierarchical index $I : \mathscr{H} \to \mathbb{R}$ satisfies INN if $I(h) = 0$ whenever $h \in \mathscr{H}$ consists of a single isolated node.

4.4.1 Extensive Population Growth (EPG)

Definition 3. A hierarchical index $I : \mathscr{H} \to \mathbb{R}$ is a *total bilateral index* if there exists a local evaluation function $g : \mathbb{Z}_+ \times \mathbb{Z}_+ \to \mathbb{R}$ such that for all $h \in \mathscr{H}$,

$$I(h) = \sum_{v \in V(h)} g\big(s_h(v), \sigma_h(v)\big).$$

To enforce that macroscopic structural change registers as a genuine *aggregate* increase, we now introduce a regularity condition specifically for total bilateral indices. Let I be a total bilateral index characterized by a local evaluation function g. We maintain the exact notation from Section 4.3.1 for the node-level coordinate shifts $\Delta s_k(v)$ and $\Delta\sigma_k(v)$, the definition of

a *macroscopic* sequence of compounds ($h'_k \in h_k \uplus h_k$), and the marginal evaluation gain $\Delta g_k(v)$.

Because total bilateral indices aggregate via an unweighted sum, we must adjust the outlier trimming term to remove the population denominator. Let $\Delta g^+_{k,(1)} \geq \Delta g^+_{k,(2)} \geq \cdots$ denote the order statistics of the multiset $\{\Delta g^+_k(v) : v \in V(h'_k)\}$, where $\Delta g^+_k(v) = \max\{\Delta g_k(v), 0\}$. For any integer $c \geq 1$, define

$$M^{\text{ext}}_k(c) = \sum_{i=1}^{c} \Delta g^+_{k,(i)},$$

which represents the aggregate structural gain attributable to the top c outlier nodes experiencing the largest positive shifts.

Condition 3 (EPG: Extensive Population Growth)**.** *A total bilateral index I satisfies* EPG *if for any macroscopic sequence of compounds ($h'_k \in h_k \uplus h_k$) with $|V(h_k)| \to \infty$ and any fixed integer $c \geq 1$,*

$$\liminf_{k\to\infty} \left[\big(I(h'_k) - 2I(h_k)\big) - M^{\text{ext}}_k(c) \right] > 0.$$

Remark 11. The EPG condition asserts that systemic structural advancement of an established population must generate a strictly positive aggregate increase even after discounting the potentially massive gains of any fixed finite set of outlier nodes. This is the extensive analogue of the trimming logic used in IPG. Note the baseline adjustment $2I(h_k)$: because total indices scale additively, the baseline volume of two disjoint copies of h_k is exactly twice the volume of h_k.

Proposition 2. *If a total bilateral index satisfies* EPG*, its local evaluation function $g : \mathbb{Z}_+ \times \mathbb{Z}_+ \to \mathbb{R}$ must be strictly increasing in both arguments. Furthermore, for any fixed value of one coordinate, the discrete marginals must satisfy*

$$\liminf_{\sigma\to\infty} \sigma \Delta_\sigma g(s,\sigma) > 0 \quad \textit{and} \quad \liminf_{s\to\infty} s \Delta_s g(s,\sigma) > 0.$$

Proof. Suppose I is a total bilateral index characterized by $g(s,\sigma)$. We evaluate the aggregate structural shifts directly using the macroscopic compound sequences $h'_m \in h_m \uplus h_m$ constructed in the proof of Proposition 1.

Step 1 (strict increase in span). Assume, toward a contradiction, that g is not strictly increasing in span. Then there exist $s, \sigma \in \mathbb{Z}_+$ such that $g(s,\sigma+1) \leq g(s,\sigma)$. Let

$\delta = g(s, \sigma + 1) - g(s, \sigma) \leq 0$.

Apply the exact compound hierarchy sequence $h'_m \in h_m \uplus h_m$ constructed in Step 1 of the proof of Proposition 1, containing $K_m = m$ targets shifting from (s, σ) to $(s, \sigma + 1)$. In the compound h'_m, the fraction of targets $K_m / |V(h'_m)|$ remains strictly bounded away from zero for all sufficiently large m, ensuring the sequence is macroscopic. By construction, all strictly positive gains are confined to the residual set $\widetilde{R}_m$. The size of this set is strictly bounded by a finite constant $c_0 = s + 2\bar{c} + 2$, which is independent of m.

Because the baseline volume of the two disjoint copies under the additive framework is exactly $2I(h_m)$, the aggregate change is

$$I(h'_m) - 2I(h_m) = \sum_{v \in V(h'_m)} \Delta g_m(v) \leq K_m \, \delta + \sum_{v \in \widetilde{R}_m} \Delta g_m^+(v).$$

Evaluating EPG at $c = c_0$, the extensive trimming term $M_m^{\text{ext}}(c_0)$ by definition sums the c_0 largest positive gains across the entire compound. Since $|\widetilde{R}_m| \leq c_0$, we have $\sum_{v \in \widetilde{R}_m} \Delta g_m^+(v) \leq M_m^{\text{ext}}(c_0)$. Subtracting $M_m^{\text{ext}}(c_0)$ thus removes at least the total positive contribution of $\widetilde{R}_m$ in the order-statistic sense. Since targets experience a gain of $\delta \leq 0$ and all nodes outside both the targets and the residual set experience nonpositive gains, we obtain

$$\big(I(h'_m) - 2I(h_m)\big) - M_m^{\text{ext}}(c_0) \leq K_m \, \delta \leq 0.$$

Taking $\liminf_{m \to \infty}$ strictly contradicts EPG. Hence $g(s, \sigma + 1) > g(s, \sigma)$ for all s, σ.

Step 2 (strict increase in depth). A symmetric argument applies. If $g(s+1, \sigma) \leq g(s, \sigma)$, we employ the sequence constructed in Step 2 of the proof of Proposition 1, wherein $K_m = m$ targets experience a unit increase in depth. All positive spillovers are confined to a residual set $\widetilde{R}'_m$ whose size is strictly bounded by a finite constant $c'_0 = \sigma + 2\bar{c} + 2$. Since $|\widetilde{R}'_m| \leq c'_0$, we have $\sum_{v \in \widetilde{R}'_m} \Delta g_m^+(v) \leq M_m^{\text{ext}}(c'_0)$. Subtracting the trimming term again removes at least the positive spillovers in the order-statistic sense, yielding a nonpositive aggregate gain and contradicting EPG. Thus $g(s + 1, \sigma) > g(s, \sigma)$.

Step 3 (decay rate of marginals). Fix s. Suppose the discrete marginals decay so rapidly in σ that $\liminf_{\sigma \to \infty} \sigma \Delta_\sigma g(s, \sigma) = 0$. Then there exists a strictly increasing sequence $\sigma_m \to \infty$ such that $\sigma_m \Delta_\sigma g(s, \sigma_m) \to 0$.

We reuse the span gadget from Step 1, placing $K_m = \sigma_m$ targets at fixed depth s and

varying span σ_m. Because the targets share the σ_m leaves, we define the baseline hierarchy size as exactly $|V(h_m)| = 2\sigma_m + s + \bar{c} + 2$. In the compound h'_m, at least $K_m = \sigma_m$ nodes experience a strictly positive span shift. The total node count is $|V(h'_m)| = 2|V(h_m)|$. As $m \to \infty$, the fraction of shifting nodes $K_m/|V(h'_m)|$ converges to $1/4$, meaning it remains strictly bounded away from zero for all sufficiently large m, ensuring the sequence is macroscopic.

By Steps 1 and 2, g is strictly increasing, so the target nodes experience strictly positive gains. The aggregate target gain is exactly $K_m \Delta_\sigma g(s, \sigma_m) = \sigma_m \Delta_\sigma g(s, \sigma_m)$, which by assumption converges to 0. As in Step 1, all positive spillovers outside the targets are confined to a finite residual set $\widetilde{R}_m$ of size at most c_0, which is independent of the growing span σ_m. Because $|\widetilde{R}_m| \leq c_0$, their total contribution is mathematically upper-bounded by the trimming term: $\sum_{v \in \widetilde{R}_m} \Delta g_m(v) \leq \sum_{v \in \widetilde{R}_m} \Delta g_m^+(v) \leq M_m^{\text{ext}}(c_0)$. Thus, subtracting the trimming term dominates the residual spillovers, leaving the aggregate gain bounded above by the target contributions:

$$\big(I(h'_m) - 2I(h_m)\big) - M_m^{\text{ext}}(c_0) \leq \sigma_m \Delta_\sigma g(s, \sigma_m).$$

Taking the limit infimum as $m \to \infty$, the right-hand side converges to 0, which strictly contradicts the positivity required by EPG. Thus, we must have $\liminf_{\sigma \to \infty} \sigma \Delta_\sigma g(s, \sigma) > 0$.

A symmetric argument using the depth gadget from Step 2, wherein the fraction of shifting nodes again converges to a positive constant, yields $\liminf_{s \to \infty} s \Delta_s g(s, \sigma) > 0$ for each fixed σ. This completes the proof. ■

4.4.2 Axiomatic characterization

Theorem 4. *A hierarchical index $I : \mathscr{H} \to \mathbb{R}$ satisfies* A, ADD, ETN, INN, *and its corresponding total bilateral representation satisfies* EPG, *if and only if there exists a graph-independent function $g : \mathbb{Z}_+ \times \mathbb{Z}_+ \to \mathbb{R}$ such that $g(0, 0) = 0$,*

$$I(h) = \sum_{v \in V(h)} g\big(s_h(v), \sigma_h(v)\big) \quad \textit{for all } h \in \mathscr{H},$$

and g satisfies the asymptotic growth condition: for every macroscopic sequence of compounds ($h'_k \in h_k \uplus h_k$) with $|V(h_k)| \to \infty$ and every integer $c \geq 1$,

$$\liminf_{k\to\infty}\left(\sum_{v\in V(h'_k)} \Delta g_k(v) - \sum_{i=1}^{c} \Delta g^{+}_{k,(i)}\right) > 0.$$

Remark 12. One might wonder why the asymptotic growth condition is not replaced by the simpler axis-based bounds identified in Proposition 2 (i.e., strict monotonicity and $\liminf x\Delta_x g > 0$), or by a strict uniform lower bound mirroring the intensive framework (Theorem 3).

First, while the marginal bounds in Proposition 2 are *necessary* consequences of EPG, they are not generally *sufficient*. Because macroscopic shifts can distribute coordinate gains across highly irregular, asymmetric graph topologies, pointwise limits along isolated axes cannot perfectly bound the aggregate trimmed growth required across all possible sequences. The growth condition quantifies over these sequences to achieve exact sufficiency.

Second, imposing a uniform strict lower bound on an unweighted sum (e.g., $\Delta g \geq m > 0$) would mathematically outlaw diminishing marginal returns, forbidding natural sublinear evaluation functions such as logarithmic or square-root growth. In an aggregate measure, it is economically plausible that adding the hundredth layer of depth contributes marginally less to the total structural volume than adding the second layer. The growth condition deliberately accommodates this decay, enforcing only the asymptotic limit: marginal gains may shrink, but they must remain collectively large enough across a macroscopic shift to strictly offset the removal of any finite number of outliers.

Proof of Theorem 4. **Necessity (⇒).** Suppose I satisfies A, ADD, ETN, INN, and its corresponding total bilateral representation satisfies EPG. By Lemma 2, the combination of A, ADD, and ETN implies that I is a total bilateral index, admitting a representation of the form

$$I(h) = \sum_{v\in V(h)} g(s_h(v), \sigma_h(v)),$$

for some real-valued local evaluation function g. By INN, $g(0,0) = 0$.

Because I satisfies EPG, for any macroscopic sequence of compounds ($h'_k \in h_k \uplus h_k$) with $|V(h_k)| \to \infty$ and any fixed integer $c \geq 1$, we have

$$\liminf_{k\to\infty}\left[I(h'_k) - 2I(h_k) - M^{\text{ext}}_k(c)\right] > 0.$$

Under the total bilateral representation, additivity over disjoint unions yields $I(h_k \cup h_k) = 2I(h_k)$ (where $\cup$ denotes the disjoint union), and the difference in the index can be expressed precisely as the sum of node-level evaluation increments:

$$I(h'_k) - 2I(h_k) = \sum_{v \in V(h'_k)} \Delta g_k(v).$$

Moreover, by definition of the extensive outlier term,

$$M_k^{\text{ext}}(c) = \sum_{i=1}^{c} \Delta g^+_{k,(i)},$$

where $\Delta g^+_{k,(1)} \geq \Delta g^+_{k,(2)} \geq \cdots$ are the order statistics of the positive gains. Substituting these identities into the EPG inequality yields exactly the asymptotic growth condition:

$$\liminf_{k \to \infty} \left(\sum_{v \in V(h'_k)} \Delta g_k(v) - \sum_{i=1}^{c} \Delta g^+_{k,(i)} \right) > 0.$$

This establishes necessity.

Sufficiency ($\Leftarrow$). Suppose I admits the displayed representation where $g(0,0) = 0$ and g satisfies the asymptotic growth condition. By the sufficiency direction of Lemma 2, any index possessing this functional dependence strictly on depth and span coordinates satisfies A, ADD, and ETN. The condition INN is immediate from $g(0,0) = 0$.

To establish EPG, consider any macroscopic sequence of compounds $(h'_k \in h_k \uplus h_k)$ with $|V(h_k)| \to \infty$ and any fixed integer $c \geq 1$. Under the total bilateral representation, we have the exact identities

$$I(h'_k) - 2I(h_k) = \sum_{v \in V(h'_k)} \Delta g_k(v) \qquad \text{and} \qquad M_k^{\text{ext}}(c) = \sum_{i=1}^{c} \Delta g^+_{k,(i)}.$$

Therefore, the expression inside the limit infimum in EPG is exactly the expression appearing in the asymptotic growth condition:

$$I(h'_k) - 2I(h_k) - M_k^{\text{ext}}(c) = \sum_{v \in V(h'_k)} \Delta g_k(v) - \sum_{i=1}^{c} \Delta g^+_{k,(i)}.$$

By the asymptotic growth condition, the limit infimum of this expression is strictly positive.

Hence,

$$\liminf_{k\to\infty}\left[I(h'_k) - 2I(h_k) - M_k^{\text{ext}}(c)\right] > 0,$$

so the index successfully leaves a persistent aggregate structural trace and satisfies EPG. This establishes sufficiency. ■

Remark 13. The role of EPG and the redundancy of unconstrained extensions. In the average framework, IPG prevents macroscopic structural gains from being mathematically diluted by the growing population denominator. While population dilution is naturally absent in the extensive setting, EPG still plays a crucial role in streamlining the framework. Because EPG guarantees that the evaluation function is strictly increasing in both coordinates (Proposition 2), it renders the standard monotonicity axioms—UE, DE, and SR—redundant for the characterization. Any index satisfying the properties of Theorem 4 naturally increases under unconstrained extensions and strictly decreases upon the removal of non-redundant edges. Finally, INN provides the essential boundary condition ($g(0,0) = 0$) that prevents the aggregate sum from conflating raw unstructured headcount with genuine hierarchical volume.

Remark 14. Admissible specifications for total indices. Because extensive indices do not suffer from population dilution, the baseline axioms alone (A, ADD, ETN, and INN) admit a vastly broader class of evaluation functions than their intensive counterparts. Absent the EPG regularity condition, any valid total index representation requires only $g(0,0) = 0$. Imposing EPG deliberately restricts this broad class by enforcing the asymptotic growth condition. While EPG accommodates decaying marginals and sublinear functions (e.g., $g(s,\sigma) = \sqrt{s} + \sqrt{\sigma}$ or logarithmic growth)—thereby allowing for natural diminishing returns—it strictly rules out asymptotically bounded functions. The bounds identified in Proposition 2 ensure that while the marginal value of structural expansion may diminish, it cannot decay faster than $1/x$, guaranteeing that the aggregate evaluation of a growing hierarchy remains capable of unbounded macroscopic growth.

Remark 15. Unilateral equivalence in the extensive framework. For any hierarchy h, the identity $\sum_v s_h(v) = \sum_v \sigma_h(v)$ holds, since both sides count ancestor–descendant pairs. Hence, any unilateral total index of the form $I(h) = \sum_v g_u(s_h(v))$ can be rewritten (up to a positive scalar multiple) as a bilateral one. Define

$$\widehat{g}(s,\sigma) = \beta g_u(s) + \alpha(\sigma - s),$$

with $\alpha, \beta > 0$. Then $\sum_v \widehat{g}(s_h(v), \sigma_h(v)) = \beta \sum_v g_u(s_h(v))$, so the global evaluation is unchanged up to the strictly positive factor β (which preserves the induced ranking), and $\widehat{g}(0,0) = 0$ whenever $g_u(0) = 0$ (which holds under INN, ensuring the axiom is preserved). Moreover, $\widehat{g}$ is strictly increasing in σ iff $\alpha > 0$, and strictly increasing in s iff $\alpha < \beta \inf_s \Delta g_u(s)$, where $\Delta g_u(s) = g_u(s+1) - g_u(s)$. Thus, if $\inf_s \Delta g_u(s) > 0$, one can choose $\alpha < \beta \inf_s \Delta g_u(s)$ to obtain a strictly increasing bilateral representative. If instead $\inf_s \Delta g_u(s) = 0$ (e.g. $g_u(s) = \sqrt{s}$), no choice of $\alpha > 0$ makes $\widehat{g}$ strictly increasing in depth; such diminishing-marginal unilateral functions are admissible under A + ADD + ETN + INN but are ruled out once EPG is imposed.

4.4.3 Non-redundancy and scope of the assumptions

The assumptions in the extensive characterization are nested by design. ADD introduces a witness f on environments, and ETN is formulated as a restriction on that witness. Moreover, EPG is defined (and evaluated) only for *total bilateral* indices, i.e., indices admitting a graph-independent representation

$$I(h) = \sum_{v \in V(h)} g\big(s_h(v), \sigma_h(v)\big)$$

obtained via A+ADD+ETN (Lemma 2). Because of this typing, strict "logical independence" of {A, ADD, ETN, INN, EPG} is not well-posed under the baseline definitions: dropping ADD removes the domain on which ETN is evaluated, and dropping ETN removes the canonical reduction that yields the universal $g(s, \sigma)$ representation needed to state EPG. Accordingly, we establish non-redundancy by providing, for each property, a witness satisfying the remaining well-scoped properties and exhibiting the pathology the dropped property is meant to exclude.

Throughout, ETN is interpreted as invariance of the index under within-environment rewiring that preserves each node's ancestor and descendant sets (hence preserves depth and span coordinates).

Dropping A. Assign fixed positive weights $\omega(v)$ based on node labels and define the label-weighted index:

$$I_{\mathrm{w}}(h) = \sum_{v \in V(h)} \omega(v)\big(s_h(v) + \sigma_h(v)\big).$$

This index satisfies ADD and INN. It satisfies ETN under the interpretation above because ETN-admissible rewirings preserve $s_h(v)$ and $\sigma_h(v)$ for each node, while $\omega(v)$ is label-fixed. It violates A because relabeling alters the weights assigned to structural positions. Since this witness does not admit a graph-independent $g(s, \sigma)$ representation, EPG is out of scope.

Dropping ADD. Consider the squared-sum index:

$$I_{\mathrm{sq}}(h) = \left(\sum_{v \in V(h)} \big(s_h(v) + \sigma_h(v) \big) \right)^2 .$$

This index satisfies A and INN. It also satisfies the index-level invariance induced by ETN (any ETN-admissible rewiring preserves each node's $s_h(v)$ and $\sigma_h(v)$ and hence preserves the inner sum). However, it violates ADD because the outer square prevents additivity over disjoint unions: $I_{\mathrm{sq}}(h_1 \cup h_2) \neq I_{\mathrm{sq}}(h_1) + I_{\mathrm{sq}}(h_2)$ in general. Since it is not in the additive class, EPG is out of scope.

Dropping ETN. Let $d_h^+(v)$ denote the out-degree of node v (number of immediate children). Define the routing-sensitive index:

$$I_{\mathrm{deg}}(h) = \sum_{v \in V(h)} \big(s_h(v) + \sigma_h(v) + d_h^+(v) \big).$$

This index satisfies A, ADD, and INN. It violates ETN: consider the three-node chain $v \to a \to b$, where $\mathrm{Desc}_h(v) = \{a, b\}$ and $d_h^+(v) = 1$. Add the redundant edge $v \to b$. Since v already reaches b via a, ancestor and descendant sets (and thus depth/span coordinates) of all nodes are unchanged, but $d_h^+(v)$ increases to 2. Hence the index changes under coordinate-preserving rewiring, contradicting ETN.

Dropping INN. Consider the baseline-shifted index:

$$I_{\mathrm{base}}(h) = \sum_{v \in V(h)} \big(s_h(v) + \sigma_h(v) + 1 \big).$$

This index satisfies A, ADD, and ETN (under the ETN interpretation above). It violates INN because a single isolated node v yields $I_{\mathrm{base}}(v) = 1 \neq 0$. It satisfies EPG because the constant +1 cancels in the expression $I(h'_k) - 2I(h_k)$, and the remaining local evaluation

$s + \sigma$ has unit discrete marginals in both coordinates, so any macroscopic coordinate growth yields macroscopic aggregate gains after trimming finitely many outliers.

Dropping EPG. Consider the span-unilateral (depth-blind) index

$$I_{\sqrt{\sigma}}(h) = \sum_{v \in V(h)} \sqrt{\sigma_h(v)}.$$

This index satisfies A, ADD, ETN, and INN: it is a node-additive sum of a function of the ETN-invariant coordinate $\sigma_h(v)$, and an isolated node has $\sigma = 0$. However, it violates EPG. Use the depth-gadget macroscopic compounds from the proof of Proposition 2, choosing the gadget with a fixed span σ_0 (e.g. $\sigma_0 = 0$), so that the set of nodes with strictly positive span spillovers is confined to a residual set of size at most c independent of k (matching EPG's fixed-c quantifier). Arrange the compounds so that a non-vanishing fraction ρ of targeted nodes gains one ancestor (strictly increasing depth) while descendant sets remain unchanged (hence $\Delta\sigma = 0$ for targets). Since the local evaluation is $g(s, \sigma) = \sqrt{\sigma}$, every targeted node has $\Delta g = 0$. After the trimming term $M_k^{\text{ext}}(c)$ removes the bounded residual positive gains, the trimmed aggregate gain equals 0, contradicting the strict positivity requirement in EPG.

4.4.4 Compliance with the axiomatic framework

To complete the analysis of the total bilateral indices characterized in Theorem 4, we summarize their compliance with the broader set of axioms considered in this paper.

Core characterization. By Theorem 4, every index in this family takes the form $I(h) = \sum_{v \in V(h)} g\big(s_h(v), \sigma_h(v)\big)$, where $g(0, 0) = 0$ and g satisfies the asymptotic growth condition required by EPG. Consequently, the family satisfies A, ADD, ETN, INN, and EPG. Furthermore, by Proposition 2, this growth condition implies that g is strictly increasing in both coordinates, which drives compliance with the remaining structural monotonicity axioms.

Subordination Removal. Strict coordinatewise monotonicity guarantees SR. Edge removal never creates reachability; any shortcut (p, i) added by Definition 1 merely preserves existing paths $(p \to j \to i)$. Thus, structural coordinates are weakly non-increasing. Redundant edge removal leaves reachability unchanged, yielding $I(h) = I(h')$. Non-redundant removal

Table 2: Characterization and axiomatic compliance of the total bilateral index family

Family characterization (Theorem 4)		**Description**
Functional form		$I(h) = \sum_{v \in V(h)} g\big(s_h(v), \sigma_h(v)\big)$
Properties of g		The local evaluation function g depends on a node's depth (s) and span (σ), with a zero-baseline condition $g(0,0) = 0$. It satisfies an asymptotic growth condition (EPG): macroscopic structural expansion yields a strictly positive aggregate evaluation gain even after trimming a bounded number of maximal positive spillovers.
Axiomatic Compliance	**Status**	**Reason**
A (Anonymity)	✓	Evaluation depends only on structural depth and span, invariant under relabeling.
ADD (Additivity)	✓	Index is defined as an unweighted sum of localized node evaluations.
ETN (Env. Topology Neutrality)	✓	Local score determined solely by ancestor/descendant counts, independent of internal routing.
INN (Isolated Node Norm.)	✓	Isolated nodes evaluate to $g(0,0) = 0$, contributing zero to the aggregate sum.
EPG (Extensive Pop. Growth)	✓	Satisfied by the growth condition in Theorem 4.
SR (Subordination Removal)	✓	Strict coordinatewise monotonicity ensures the index strictly decreases upon non-redundant edge removal.
UE* / DE* (Relaxed Extensions)	✓	Any partial attachment strictly increases the unweighted aggregate sum.
UE / DE (Vertical Extensions)	✓	Aggregate sum strictly increases; no population dilution effect for appended nodes.
QS (Quadratic Separability)	×	Incompatible with extensive summation; value scales linearly rather than inversely with replication.
AS (Additive Separability)	×	Incompatible with extensive summation; index sum scales linearly with replication rather than averaging.

strictly decreases at least one node's depth or span (e.g., a subordinate loses a supervisor). Since g is strictly increasing, the aggregate sum strictly decreases.

Relaxed extension axioms. This family satisfies UE* and DE*. Any partial extension $h^* \in h_o \uplus h$ adds at least one cross-edge, strictly increasing the depth or span of at least one node while decreasing none. Since g is strictly increasing, the compound sum strictly exceeds that of the disjoint union $h_o \cup h$. By ADD, $I(h_o \cup h) = I(h_o) + I(h)$. Because $g(0,0) = 0$ and g is strictly increasing, $g(s,\sigma) \geq 0$ for all $(s,\sigma) \in \mathbb{Z}_+^2$, hence $I(\cdot) \geq 0$. Thus, we obtain $I(h^*) > I(h_o) + I(h) \geq \max\{I(h_o), I(h)\}$. The required strict inequalities therefore hold unconditionally, bypassing the weak dominance precondition needed in the intensive setting.

Unrestricted extension axioms. Unlike average bilateral indices, this family fully satisfies UE and DE. In any complete extension $h^* \in h_o \oplus h$ (resp. $h^* \in h \boxplus h_o$), every leaf (resp. root) of h_o connects to h. Consequently, every appended node acquires strictly positive coordinates ($(s,\sigma) \neq (0,0)$) and contributes strictly positively to the sum. Since pre-existing nodes in h retain all prior reachability, their evaluations weakly increase. Because the extensive sum lacks a population denominator, appending h_o strictly increases the aggregate score ($I(h^*) > I(h)$) without dilution, regardless of h_o's size or flatness.

Quadratic Separability and Additive Separability. QS imposes a quadratic density normalization, and AS imposes intensive population averaging (yielding replication invariance, $I(h^k) = I(h)$). Both are mathematically incompatible with extensive summation. Under ADD, the index of a replicated hierarchy scales additively ($I(h^k) = kI(h)$), precluding both QS and AS.

4.5 A class of bilateral concentration indices

The preceding sections characterized average and total indices by applying linear (AS) and absolute (ADD) separability axioms. We now complete the structural trichotomy by characterizing the family of indices derived from Quadratic Separability (QS).

Indices in this class evaluate hierarchy through the lens of structural density. Rather than asking "what is the average hierarchical experience?" or "how much total hierarchy exists?", these indices ask "how concentrated is the hierarchy relative to the theoretical capacity of the population?"

When ETN is combined with the quadratic population-density structure of QS and the node-label independence of Anonymity (A), it constrains the functional form of an index, forcing it to evaluate organizations exclusively through the lens of individual depth and span. Throughout this subsection, ETN is evaluated relative to the separability witness provided by QS: that is, when I satisfies A and QS with witness f, ETN requires $f(N(v,h))$ to be invariant to environment-edge restructurings that preserve each node's absolute sets of supervisors and subordinates. We formalize this structural reduction below.

Lemma 3. *A hierarchical index $I : \mathscr{H} \to \mathbb{R}$ satisfies A, QS, and ETN if and only if there exists a function $g : \mathbb{Z}_+ \times \mathbb{Z}_+ \to \mathbb{R}$ such that for every $h \in \mathscr{H}$,*

$$I(h) = \frac{1}{|V(h)|^2} \sum_{v \in V(h)} g(s_h(v), \sigma_h(v)). \tag{4}$$

The proof is entirely analogous to that of Lemma 1 and therefore omitted.

4.5.1 Concentration Population Growth (CPG)

Definition 4. A hierarchical index $I : \mathscr{H} \to \mathbb{R}$ is a *bilateral concentration index* if there exists a local evaluation function $g : \mathbb{Z}_+ \times \mathbb{Z}_+ \to \mathbb{R}$ such that for all $h \in \mathscr{H}$,

$$I(h) = \frac{1}{|V(h)|^2} \sum_{v \in V(h)} g\big(s_h(v), \sigma_h(v)\big).$$

To ensure the local evaluation function $g(s, \sigma)$ correctly rewards hierarchical depth and span, we require a regularity condition analogous to IPG.

However, we cannot apply IPG directly to concentration indices. A macroscopic sequence of compounds injects new structure proportional to the population size. Because a concentration index penalizes population quadratically, this denominator growth consistently overwhelms structural additions. Thus, as a general rule for density metrics where the numerator grows at most linearly in $|V(h)|$ along macroscopic compounding, the raw difference $I(h'_k) - I(h_k)$ typically becomes negative—quadratic denominator growth can easily dominate the linear structural additions—so applying IPG directly is inappropriate.

To evaluate structural monotonicity without density dilution, we multiply the index by the squared population to recover the absolute structural volume. Dividing the marginal change in this volume by the base population yields the *per-capita structural accumulation*, mirroring the mechanics of IPG.

We formalize this using the comparative framework and notation from Section 4.3.1. Form h'_k from two node-disjoint isomorphic copies $h_k^{\uparrow}$ and $h_k^{\downarrow}$ of a baseline hierarchy h_k, connected by some cross-edges (i.e., $h'_k \in h_k \uplus h_k$). Thus, $|V(h'_k)| = 2|V(h_k)|$ exactly. Fix isomorphisms $\phi_k^{\uparrow} : h_k^{\uparrow} \to h_k$ and $\phi_k^{\downarrow} : h_k^{\downarrow} \to h_k$. For each $v \in V(h'_k)$, let $v_0 \in V(h_k)$ denote its baseline counterpart (i.e., $v_0 = \phi_k^{\uparrow}(v)$ if $v \in V(h_k^{\uparrow})$, and $v_0 = \phi_k^{\downarrow}(v)$ if $v \in V(h_k^{\downarrow})$). We define the coordinate shifts

$$\Delta s_k(v) = s_{h'_k}(v) - s_{h_k}(v_0) \quad \text{and} \quad \Delta\sigma_k(v) = \sigma_{h'_k}(v) - \sigma_{h_k}(v_0),$$

alongside the marginal evaluation gain

$$\Delta g_k(v) = g\big(s_{h'_k}(v), \sigma_{h'_k}(v)\big) - g\big(s_{h_k}(v_0), \sigma_{h_k}(v_0)\big).$$

To isolate systemic growth, we define $\Delta g_k^+(v) = \max\{\Delta g_k(v), 0\}$. Letting $\Delta g_{(1)}^+ \geq \Delta g_{(2)}^+ \geq \cdots$ be the order statistics of the multiset $\{\Delta g_k^+(v) : v \in V(h'_k)\}$, we define the absolute outlier accumulation for any integer $c \geq 1$ as:

$$M_k^{\text{con}}(c) = \sum_{i=1}^{c} \Delta g_{(i)}^+.$$

Condition 4 (CPG: Concentration Population Growth)**.** *A bilateral concentration index I satisfies CPG if, for any macroscopic sequence $(h'_k \in h_k \uplus h_k)$ formed from a baseline hierarchy h_k such that $|V(h_k)| \to \infty$, the per-capita structural accumulation strictly outpaces the contribution of any finite set of outlier nodes. Specifically, for any fixed integer $c \geq 1$:*

$$\liminf_{k\to\infty} \frac{1}{|V(h_k)|} \left[|V(h'_k)|^2 I(h'_k) - 2|V(h_k)|^2 I(h_k) - M_k^{\text{con}}(c)\right] > 0.$$

Remark 16. While QS imposes a density interpretation, it cannot guarantee that the evaluation function g strictly rewards structural growth. Whenever an organization expands sparsely, the quadratic population penalty naturally drives the overall index toward zero. This dilution obscures the underlying mechanics: a declining index could reflect either a genuine drop in structural density or a saturated evaluation function g that has simply stopped assigning value to new hierarchy. CPG resolves this ambiguity by stripping away the quadratic denominator, ensuring that macroscopic structural deepening leaves a strictly positive trace on the per-capita structural accumulation, regardless of how the overarching

density shifts.

This density-adjusted condition yields the direct analogue of Proposition 1 for bilateral concentration indices.

Proposition 3. *If a bilateral concentration index satisfies CPG, its local evaluation function $g : \mathbb{Z}_+ \times \mathbb{Z}_+ \to \mathbb{R}$ must be strictly increasing in both arguments and, for any fixed value of one coordinate, have discrete marginals strictly bounded away from zero in the other argument.*

Proof. By Lemma 3, $I(h) = \frac{1}{|V(h)|^2} \sum_{v \in V(h)} g(s_h(v), \sigma_h(v))$. Substituting this into the bracketed term of CPG cancels the quadratic denominators, isolating the absolute aggregate change. Because $h'_k \in h_k \uplus h_k$ comprises two disjoint copies of h_k, the baseline structural mass is $2 \sum_{v \in V(h_k)} g(s_{h_k}(v), \sigma_{h_k}(v))$. The bracketed term thus expands to:

$$|V(h'_k)|^2 I(h'_k) - 2|V(h_k)|^2 I(h_k) - M_k^{\text{con}}(c) \;=\; \sum_{v \in V(h'_k)} \Delta g_k(v) - \sum_{i=1}^{c} \Delta g^+_{(i)}.$$

Since $|V(h'_k)| = 2|V(h_k)|$, dividing this absolute difference by $|V(h_k)|$ yields exactly:

$$\frac{2}{|V(h'_k)|} \left[\sum_{v \in V(h'_k)} \Delta g_k(v) - \sum_{i=1}^{c} \Delta g^+_{(i)} \right].$$

Therefore, the CPG inequality is mathematically identical to:

$$\liminf_{k \to \infty} \; 2 \left[I^{\text{avg}}(h'_k) - I^{\text{avg}}(h_k) - M_k^{\text{avg}}(c) \right] > 0,$$

where $I^{\text{avg}}(h) = \frac{1}{|V(h)|} \sum_{v \in V(h)} g(s_h(v), \sigma_h(v))$ is the bilateral average index defined by the identical evaluation function g. Because the average index is invariant to replication, $I^{\text{avg}}(h_k)$ is exactly equal to the average index of the disjoint union $h_k^{\uparrow} \cup h_k^{\downarrow}$. Moreover, $M_k^{\text{avg}}(c) = \frac{1}{|V(h'_k)|} \sum_{i=1}^{c} \Delta g^+_{(i)}$ is exactly the intensive outlier trimming term evaluated over the identical macroscopic sequence.

Because CPG imposes an algebraic condition identical to IPG (up to a strictly positive scalar multiplier) over the same class of macroscopic sequences, g satisfies CPG if and only if it satisfies IPG. The necessity of strict monotonicity and bounded marginals therefore follows immediately from the gadget constructions in the proof of Proposition 1. ∎

4.5.2 Axiomatic characterization

With the foundational structure and regularity conditions established, we can now state the main representation theorem for this family. This result parallels the characterization of average bilateral indices (Theorem 3), demonstrating that replacing the separability axiom completely redefines the macroscopic scaling of the index without altering the localized evaluation of hierarchy.

Lemma 4 (CPG–IPG Equivalence)**.** *Under the bilateral concentration representation* $I(h) = \frac{1}{|V(h)|^2}\sum_{v\in V(h)} g(s_h(v), \sigma_h(v))$*, the index satisfies* CPG *if and only if the corresponding average bilateral index,*

$$I^{\text{avg}}(h) = \frac{1}{|V(h)|}\sum_{v\in V(h)} g\big(s_h(v), \sigma_h(v)\big),$$

satisfies IPG*, where* IPG *is evaluated over the exact same macroscopic sequences of compounds* $h'_k \in h_k \uplus h_k$ *used in* CPG*.*

Theorem 5. *A hierarchical index* $I : \mathscr{H} \to \mathbb{R}$ *satisfies* A*,* QS*,* ETN*, and* CPG *if and only if there exists a function* $g : \mathbb{Z}_+ \times \mathbb{Z}_+ \to \mathbb{R}$ *such that for all* $h \in \mathscr{H}$*,*

$$I(h) = \frac{1}{|V(h)|^2}\sum_{v\in V(h)} g\big(s_h(v), \sigma_h(v)\big),$$

and g *has discrete marginals strictly bounded away from zero: there exists a constant* $m > 0$ *such that for all* $s, \sigma \in \mathbb{Z}_+$*,*

$$m \le \Delta_s g(s,\sigma) = g(s+1,\sigma) - g(s,\sigma) \quad \textit{and} \quad m \le \Delta_\sigma g(s,\sigma) = g(s,\sigma+1) - g(s,\sigma).$$

Proof. By Lemma 3, I satisfies A, QS, and ETN if and only if it admits the stated representation $I(h) = \frac{1}{|V(h)|^2}\sum_{v\in V(h)} g(s_h(v), \sigma_h(v))$ for some local evaluation function g.

By the equivalence established in Lemma 4, this representation satisfies CPG if and only if the corresponding bilateral average index defined by the exact same evaluation function g satisfies IPG. By Theorem 3, this holds if and only if g has discrete marginals uniformly bounded below by a strictly positive constant $m > 0$. The result follows immediately. ■

Remark 17. Admissible specifications and sparse decay. Because CPG imposes an algebraic constraint identical to IPG on the local evaluation function g, the set of admissible specifications is identical to that described in Remark 9. Specifically, CPG strictly rules out

saturated (bounded) functions and sublinear unbounded growth, while permitting explosive superlinear curvature.

However, the macroscopic behavior of the overall concentration index $I(h)$ differs fundamentally from an average index. In standard, sparse organizational structures (such as trees or broad pyramids), the total volume of hierarchical relationships (i.e., the total number of ancestor–descendant pairs) grows much slower than the squared population. Consequently, dividing this structural accumulation by the theoretical capacity $|V(h)|^2$ causes the concentration index to asymptotically decay toward zero as the organization expands. This sparse decay is not a violation of CPG; rather, it is the defining mathematical signature of a density metric. CPG ensures the local evaluation function g strictly rewards the structure that actually exists, while the quadratic denominator legitimately penalizes the vast structural voids inherent to expanding sparse networks.

Finally, it is important to distinguish the permissible curvature of the local evaluation function g from transformations of the global index. While g itself may exhibit explosive non-linear curvature, QS is preserved only under positive scalar multiplication (within the class of indices admitting a graph-independent local evaluation function). Applying any strictly increasing non-linear transformation to the final aggregate index (e.g., $J(h) = T(I(h))$), or even adding a constant, destroys the separable summation structure required by QS.

Remark 18. Equivalence of unilateral and bilateral density specifications. Just as in the intensive framework (Remark 10), the strict coordinatewise monotonicity required by Theorem 5 does not exclude unilateral density indices (e.g., $g(s, \sigma) = f(s)$). Because the network identity $\sum_{v \in V(h)} s_h(v) = \sum_{v \in V(h)} \sigma_h(v)$ holds, an equivalent bilateral representation can be constructed. A convenient recasting is obtained via the zero-sum structural shift:

$$\hat{g}(s, \sigma) = \beta f(s) + \alpha(\sigma - s), \quad \text{with } \alpha, \beta > 0.$$

A sufficient condition for choosing $\alpha > 0$ so that $\hat{g}$ has discrete marginals strictly bounded away from zero in both coordinates is $\inf_s [f(s+1) - f(s)] > 0$; in that case, one may pick $0 < \alpha < \beta \inf_s [f(s+1) - f(s)]$, yielding $\Delta_s \hat{g}(s, \sigma) = \beta[f(s+1) - f(s)] - \alpha \geq m > 0$ and $\Delta_\sigma \hat{g}(s, \sigma) = \alpha \geq m > 0$, where $m = \min\{\alpha, \beta \inf_s [f(s+1) - f(s)] - \alpha\} > 0$. Note that unlike the average index transformation, this shift cannot include an additive constant, as dividing a constant by $|V(h)|^2$ would introduce a size-dependent distortion. This proportional transformation yields an equivalent index $\hat{I}(h) = \beta I(h)$ that perfectly preserves both the ordinal ranking of hierarchies and QS. Moreover, under CPG, a unilateral specification must

satisfy $\inf_s[f(s+1) - f(s)] > 0$ (otherwise the corresponding average index would violate IPG), meaning this recoding applies universally. Thus, the bilateral concentration framework natively encompasses all CPG-admissible unilateral density indices.

4.5.3 Non-redundancy and scope of the assumptions

As with the average bilateral indices (Section 4.3.3), the assumptions in Theorem 5 are nested by design. QS introduces a witness f on environments, and ETN is formulated as a restriction on that witness. Moreover, CPG is defined strictly for bilateral concentration indices. When we say CPG is formally out of scope, we mean the index does not belong to the bilateral concentration class to which CPG is defined to apply (i.e., it does not admit the required graph-independent $g(s, \sigma)$ representation). We establish non-redundancy by providing indices that satisfy the remaining well-scoped assumptions while exhibiting the excluded pathology. Throughout, a "macroscopic sequence of compounds" designates a sequence in which a non-vanishing fraction of nodes experiences positive coordinate shifts.

Dropping A. Assign weights $w(v) = 2$ if the label of v is even and 1 otherwise. The index

$$I_{\mathrm{w}}(h) = \frac{1}{|V(h)|^2} \sum_{v \in V(h)} w(v)\big(s_h(v) + \sigma_h(v)\big)$$

satisfies QS (with witness $f(N(v,h)) = w(v)\big(s_h(v) + \sigma_h(v)\big)$) and ETN, but violates A because relabeling that swaps an even-labeled node with an odd-labeled one changes the index. Because CPG is defined only for bilateral concentration indices (which automatically satisfy A), it is out of scope here.

Dropping QS. The baseline average bilateral index

$$I_{\mathrm{avg}}(h) = \frac{1}{|V(h)|} \sum_{v \in V(h)} \big(s_h(v) + \sigma_h(v)\big)$$

depends only on coordinates, trivially satisfying A and ETN. However, it violates QS. If it satisfied QS, replicating a non-flat hierarchy h (equivalently, one with $I_{\mathrm{avg}}(h) > 0$) exactly k times would force the index to scale by a factor of $1/k$; instead, I_{avg} remains strictly constant, since $I_{\mathrm{avg}}(h^k) = \frac{1}{k|V(h)|} \cdot k \sum_{v \in V(h)} \big(s_h(v) + \sigma_h(v)\big) = I_{\mathrm{avg}}(h)$. CPG is out of scope.

Dropping ETN. As in Section 4.3.3, let $m_h(v)$ be the number of edges strictly internal to a node's hierarchical environment. The routing-sensitive index

$$I_{\mathrm{rt}}(h) = \frac{1}{|V(h)|^2} \sum_{v \in V(h)} \big(s_h(v) + \sigma_h(v) + m_h(v)\big)$$

satisfies A and QS (with witness $f(N(v,h)) = s_h(v) + \sigma_h(v) + m_h(v)$). However, it violates ETN because internal rewiring alters $m_h(v)$ without changing ancestor or descendant sets. For example, if v has two incomparable root ancestors a_1 and a_2, adding the edge $a_1 \to a_2$ preserves $\mathrm{Anc}_h(v)$ and $\mathrm{Desc}_h(v)$ while keeping the environment acyclic, but it strictly increases $m_h(v)$ by 1. CPG is out of scope.

Dropping CPG. Consider the saturating density index:

$$I_{\mathrm{sat}}(h) = \frac{1}{|V(h)|^2} \sum_{v \in V(h)} \left(2 - \frac{1}{s_h(v)+1} - \frac{1}{\sigma_h(v)+1}\right)$$

It satisfies A, QS, and ETN. However, as coordinates grow large, the unit marginals ($\Delta_s g(s,\sigma) = (s+1)^{-1}(s+2)^{-1}$ and $\Delta_\sigma g(s,\sigma) = (\sigma+1)^{-1}(\sigma+2)^{-1}$) vanish. Under CPG, multiplying $I(h)$ by $|V(h)|^2$ strictly cancels the quadratic denominator. Indeed, $|V(h'_k)|^2 I(h'_k) = \sum_{v \in V(h'_k)} g\big(s_{h'_k}(v), \sigma_{h'_k}(v)\big)$, while $2|V(h_k)|^2 I(h_k) = \sum_{v \in V(h_k^\uparrow) \cup V(h_k^\downarrow)} g\big(s_{h_k}(v_0), \sigma_{h_k}(v_0)\big)$, so the bracketed term in CPG equals $\sum_{v \in V(h'_k)} \Delta g_k(v) - M_k^{\mathrm{con}}(c)$. For any $\varepsilon > 0$ and any fixed integer $c \geq 1$ (matching the parameter in CPG), one can construct a macroscopic sequence (using target-and-trunk constructions from previous proofs, which explicitly generate a macroscopic set of targeted nodes alongside a uniformly bounded spillover set) in which targeted nodes are placed at sufficiently large coordinates such that their unit marginals are at most ε. Moreover, the construction ensures each targeted node experiences at most a unit coordinate shift (e.g., $\Delta s_k(v) + \Delta\sigma_k(v) = 1$). Because compounding only adds edges, coordinates weakly increase for all nodes; coupled with a strictly increasing g, this ensures all marginal gains are non-negative ($\Delta g_k(v) \geq 0$). Consequently, each targeted gain equals either $\Delta_s g(s,\sigma)$ or $\Delta_\sigma g(s,\sigma)$, both of which are $\leq \varepsilon$ by construction. Furthermore, any positive-gain spillovers outside the targeted set are confined to a residual set of at most c nodes and are therefore fully removed by $M_k^{\mathrm{con}}(c)$. All other nodes outside the targeted and residual sets experience no coordinate shifts ($\Delta s_k(v) = \Delta\sigma_k(v) = 0$) and therefore contribute zero marginal gain. Because the remaining positive gains belong exclusively to the targeted

nodes (which constitute a fraction ρ of the $|V(h'_k)|$ total nodes), the trimmed per-capita sum is strictly bounded for all sufficiently large k:

$$\frac{1}{|V(h_k)|}\left[\sum_{v\in V(h'_k)} \Delta g_k(v) - M_k^{\mathrm{con}}(c)\right] \le \frac{\rho|V(h'_k)|\varepsilon}{|V(h_k)|} = 2\rho\varepsilon$$

(noting that $|V(h'_k)| = 2|V(h_k)|$). Consequently, the limit infimum is at most $2\rho\varepsilon$. Since $\varepsilon > 0$ is arbitrary, the strict positivity required by CPG fails.

4.5.4 Compliance with the axiomatic framework

To complete the analysis of the bilateral concentration indices characterized in Theorem 5, we summarize their compliance with the broader set of axioms considered in this paper.

Core characterization. By Theorem 5, every index in this family takes the form

$$I(h) \;=\; \frac{1}{|V(h)|^2} \sum_{v\in V(h)} g\big(s_h(v), \sigma_h(v)\big),$$

where g is strictly increasing in each argument. Consequently, the family naturally captures structural density and satisfies A, QS, ETN, and the regularity condition CPG.

Subordination Removal. By a structural argument identical to that for average indices (Section 4.3.4), the strict coordinatewise monotonicity of g guarantees SR. Edge removal never creates new reachability, meaning depth and span coordinates are weakly non-increasing. If the removed edge is redundant, the reachability relation is unchanged, so all ancestor/descendant sets (hence all $s_h(\cdot)$ and $\sigma_h(\cdot)$) remain unchanged and therefore $I(h) = I(h')$, satisfying the weak part of SR. Conversely, if the removed edge is non-redundant, at least one ancestor–descendant reachability relation is lost, so some node's ancestor set or descendant set strictly shrinks, implying a strict decrease in $s_h(\cdot)$ or $\sigma_h(\cdot)$. Because the population denominator $|V(h)|^2$ remains absolutely constant during edge removal, this coordinate reduction mathematically forces the aggregate index to strictly decrease.

Extension axioms. As established by Theorem 1 and Theorem 2, under the imposed regularity hypotheses, quadratic population normalizations (QS) are fundamentally incompatible

Table 3: Characterization and axiomatic compliance of the bilateral concentration index family

Family characterization (Theorem 5)		**Description**
Functional form		$I(h) = \frac{1}{\lvert V(h)\rvert^2} \sum_{v \in V(h)} g\big(s_h(v), \sigma_h(v)\big)$
Properties of g		The local evaluation function g depends on a node's depth (s) and span (σ). It is strictly increasing in both dimensions, meaning any unit increase in depth or span yields a strict evaluation gain. Specifically, CPG enforces that macroscopic compounding yields strictly positive rescaled per-capita structural accumulation after trimming finitely many outliers, which implies the discrete marginals of g are uniformly bounded away from zero.
Axiomatic Compliance	**Status**	**Reason**
A (Anonymity)	✓	Evaluation depends only on structural depth and span, invariant under relabeling.
QS (Quadratic Separability)	✓	Index evaluates global structural density via a quadratic population normalization.
ETN (Env. Topology Neutrality)	✓	Local score determined solely by ancestor/descendant counts, independent of internal routing.
CPG (Concentration Pop. Growth)	✓	By Theorem 5, CPG implies discrete marginals of g are uniformly bounded away from zero.
SR (Subordination Removal)	✓	Strict coordinatewise monotonicity ensures index weakly/strictly decreases upon edge removal.
UE* / DE* (Relaxed Extensions)	×†	Under regularity (Theorem 2), there exists an admissible extension that strictly decreases I, violating the required increase.
UE / DE (Vertical Extensions)	×†	Under regularity (Theorem 1), there exists an admissible extension that strictly decreases I, violating the required increase.
AS (Additive Separability)	×	Incompatible with density averaging; replication dilutes the measure rather than preserving it.
ADD (Additivity)	×	Incompatible with density averaging; index sum does not scale extensively.

†Under the regularity hypotheses of Theorem 1 and Theorem 2.

with vertical extensions. Density metrics inherently penalize population growth by squaring the population size in the denominator. Appending standard tree-like extensions introduces a number of new hierarchical relationships that scales only linearly with the population size. This quadratic dilution effect mathematically dominates the linear marginal structural gains induced by the linking edges. Consequently, by Theorem 1 (resp. Theorem 2), for any regular QS index there exists an admissible unrestricted (resp. partial) extension that strictly decreases the overall index, directly contradicting the strict increase required by UE and DE (resp. UE* and DE*). To satisfy strict extension monotonicity, a density metric would require sufficiently dense (quadratic-order) structural growth to stave off the dilution effects of population expansion.

Additivity and Additive Separability. The bilateral concentration family inherently violates both ADD and AS. ADD requires absolute extensive volume scaling, and AS requires per-capita scaling. In stark contrast, under QS, replicating a non-flat hierarchy h (with $I(h) > 0$) exactly k times fragments the concentration of power, strictly dropping the index to $\frac{1}{k}I(h)$ and thereby satisfying the Concentration Replication Principle (CRP). Thus, any non-trivial index satisfying QS necessarily violates AS and ADD, completing our map of the three fundamental paradigms of structural measurement.

The impossibility of dense extensions. At first glance, the violation of all extension axioms by the bilateral concentration family may appear as a mathematical deficiency. In reality, it represents a fundamental impossibility frontier in structural measurement. Extension axioms evaluate the aggregation of new organizational layers. However, standard organizational expansion is inherently sparse; adding a new tier of management or a massive flat base introduces a number of new hierarchical relationships that scales linearly with the population size, while the capacity for cross-organizational reporting grows quadratically. Thus, any genuine measure of global density must mathematically penalize standard structural extensions. This reveals a profound trade-off in organizational theory: one cannot simultaneously conceptualize hierarchy as a strictly accumulating mass (which demands the satisfaction of structural extensions) and as a global concentration of power (which demands their violation). The axiomatic trichotomy presented here strictly forces the analyst to choose their theoretical paradigm.

5 Hierarchical indices from the literature

In this section, we examine several hierarchical indices from the literature through the lens of our axiomatic framework. Our goal is not to rederive each measure in full, but to identify which axioms they satisfy and which they violate.

Summary of axiomatic compliance. Table 4 synthesizes the compliance of each literature index with our axiomatic framework.

Table 4: Axiomatic compliance of hierarchical indices from the literature

Axiom	**Avg. Depth** (*a*)	**GRC** (*b*)	**Conn.** (*c*)	**Graph Hier.** (*c*)	**Graph Eff.** (*c*)	**LUB** (*c*)
A	✓	✓	✓	✓†	✓	✓
AS	✓	×	×	×	×	×
ADD	×	×	×	×	×	×
QS	×	×	×	×	×	×
IRP	✓	×	×	✓†	✓	✓
ERP	×	×	×	×	×	×
CRP	×	✓	✓	×	×	×
SR	✓	×	×	×	×	×
UE/DE	×	×	×	×	×	×
UE*/DE*	✓	×	×	×	×	×
INN	✓	undef.	undef.	undef.	undef.	undef.

Notes: ✓ denotes satisfaction; × denotes violation; "undef." indicates the axiom cannot be evaluated. †Satisfied vacuously (index is constant on DAGs). The average depth index additionally satisfies conditions ETN and IPG.
Sources: (*a*) Carbonell-Nicolau (2025b); (*b*) Mones et al. (2012); (*c*) Krackhardt (1994).

5.1 Average depth

A hierarchical index measuring *average depth* is defined by

$$I_s(h) = \frac{1}{|V(h)|} \sum_{v \in V(h)} s_h(v),$$

where $s_h(v)$ is the number of (direct or indirect) supervisors (ancestors) of v in h (Carbonell-Nicolau, 2025b). Intuitively, I_s is the expected number of superiors faced by a uniformly random individual.

Location in our framework. The index I_s is a *unilateral* member of the *average bilateral* family characterized in Theorem 3, via the unilateral–bilateral equivalence in Remark 10. In particular, I_s is an *additively separable population average* with primitive local evaluation $g(s, \sigma) = s$.

Although this local evaluation depends only on depth, I_s admits an *aggregate-equivalent* strictly bilateral representation. Indeed, for every hierarchy h,

$$\sum_{v \in V(h)} s_h(v) = \sum_{v \in V(h)} \sigma_h(v),$$

so that

$$I_s(h) = \frac{1}{|V(h)|} \sum_{v \in V(h)} s_h(v) = \frac{1}{|V(h)|} \sum_{v \in V(h)} \frac{1}{2}\big(s_h(v) + \sigma_h(v)\big).$$

Thus I_s belongs to the average-bilateral family in the sense of Theorem 3, even though its primitive local evaluation is unilateral.

Moreover, $\hat{g}(s, \sigma) = \frac{1}{2}(s + \sigma)$ provides a symmetrized bilateral representation of the primitive evaluation, which is enough to verify A, AS, ETN, and IPG.

Remark 19. At first glance, one might suspect that I_s violates the intensive growth condition IPG because its primitive local evaluation $g(s, \sigma) = s$ has zero marginal with respect to span. However, the identity

$$\sum_{v \in V(h)} s_h(v) = \sum_{v \in V(h)} \sigma_h(v)$$

implies that aggregate span and aggregate depth move in lockstep on every hierarchy. Hence the average-depth index can be rewritten as a bilateral average with local evaluation $\hat{g}(s, \sigma) = \frac{1}{2}(s + \sigma)$, making the dependence on both dimensions explicit at the aggregate level. This is the relevant representation for IPG, and it shows that I_s satisfies that condition.

Replication principles. Because I_s is a population average, it satisfies IRP: $I_s(h^k) = I_s(h)$ for all $k \geq 1$. It therefore violates ERP (the extensive view), since $I_s(h^k)$ does not strictly increase with k whenever $E(h) \neq \varnothing$; and it violates CRP (the concentration view), since $I_s(h^k)$ does not strictly decrease with k whenever $E(h) \neq \varnothing$.

Monotonicity and extension axioms. Because depth weakly decreases under subordination removal, I_s satisfies SR, with strict inequality whenever a non-redundant edge is removed.

Moreover, as an average-bilateral index, it satisfies the relaxed extension axioms UE* and DE*. By contrast, it violates the unrestricted extension axioms UE and DE by Theorem 1.

Additivity, normalization, and separability. The index satisfies INN (an isolated node has depth 0 and hence contributes 0). It violates ADD by design, since it is an intensive average rather than an extensive sum. Finally, it violates QS, reflecting that it is not a concentration/density measure.

Proposition 4. *The average depth index I_s satisfies* A, AS, ETN, IPG, INN, SR, UE*, DE*, *and* IRP, *and violates* ADD, QS, UE, DE, ERP, *and* CRP.

5.2 Global reaching centrality

The *global reaching centrality* (GRC) index of Mones et al. (2012) is

$$\mathrm{GRC}(h) = \frac{1}{N-1} \sum_{v \in V(h)} \left(C_R^{\max} - C_R(v)\right), \qquad C_R^{\max} = \max_{u \in V(h)} C_R(u),$$

where $N = |V(h)|$ and $C_R(v) = \frac{|\mathrm{Desc}_h(v)|}{N-1}$ is the fraction of *other* nodes reachable from v. The index is based on the concentration of downward reach, not on upward depth.

Location in our framework. Because the local term $C_R^{\max} - C_R(v)$ depends on the global maximum $C_R^{\max}$, GRC is not a member of the bilateral families characterized in our framework. Its reliance on a globally defined reference point makes it incompatible with the separability paradigms. It does, however, satisfy anonymity (A).

Replication principles. Replication changes reach fractions because each copy is disconnected: for any node, the reachable set is unchanged in absolute size but is measured against a larger population. Thus, for the k-fold replication h^k,

$$C_R^{(h^k)}(v) = \frac{|\mathrm{Desc}_h(v)|}{kN-1} = \frac{N-1}{kN-1} C_R^{(h)}(v),$$

and therefore

$$\mathrm{GRC}(h^k) = \frac{k(N-1)^2}{(kN-1)^2} \mathrm{GRC}(h).$$

Since $\frac{k(N-1)^2}{(kN-1)^2} < 1$ for $k > 1$ and $N \geq 2$, GRC strictly decreases under replication whenever $E(h) \neq \varnothing$ (hence $N \geq 2$). It therefore satisfies CRP and violates IRP and ERP.

Monotonicity and extension axioms. GRC's global normalization creates non-monotonicities: it violates SR, UE, DE, UE*, and DE*.

Concrete violation of SR. Consider the three-node chain $1 \to 2 \to 3$. Here $N = 3$, $C_R(1) = 1, C_R(2) = \frac{1}{2}, C_R(3) = 0$, so $C_R^{\max} = 1$ and $\mathrm{GRC}(h) = \frac{1}{2}\left[(1-1)+(1-\frac{1}{2})+(1-0)\right] = \frac{3}{4}$. Removing the non-redundant edge $(2,3)$ and applying the SR shortcutting rule adds $(1,3)$, yielding edges $1 \to 2$ and $1 \to 3$. Now $C_R(1) = 1$, $C_R(2) = C_R(3) = 0$, so $C_R^{\max} = 1$ and $\mathrm{GRC}(h') = \frac{1}{2}\left[(1-1)+(1-0)+(1-0)\right] = 1 > \frac{3}{4}$. Thus GRC strictly *increases* after a non-redundant edge removal, violating SR.

Concrete violation of UE. Let h be the three-node chain $1 \to 2 \to 3$ with $\mathrm{GRC}(h) = \frac{3}{4}$ as above. Let h_o be a single isolated node ℓ, and form $h^* \in h_o \oplus h$ by adding the edge $\ell \to 1$, yielding the four-node chain $\ell \to 1 \to 2 \to 3$. Now $N = 4$ and $N - 1 = 3$. Descendant counts: $|\mathrm{Desc}_{h^*}(\ell)| = 3$, $|\mathrm{Desc}_{h^*}(1)| = 2$, $|\mathrm{Desc}_{h^*}(2)| = 1$, $|\mathrm{Desc}_{h^*}(3)| = 0$, so $C_R(\ell) = 1$, $C_R(1) = \frac{2}{3}$, $C_R(2) = \frac{1}{3}$, $C_R(3) = 0$, and $C_R^{\max} = 1$. Hence

$$\mathrm{GRC}(h^*) = \frac{1}{3}\left[(1-1) + \left(1-\tfrac{2}{3}\right) + \left(1-\tfrac{1}{3}\right) + (1-0)\right] = \frac{2}{3} < \frac{3}{4} = \mathrm{GRC}(h),$$

so an admissible upward extension *decreases* the index, violating UE.

Concrete violation of DE. Let h be the three-node chain $1 \to 2 \to 3$ with $\mathrm{GRC}(h) = \frac{3}{4}$. Let h_o be a single isolated node ℓ, and form $h^* \in h \boxplus h_o$ by adding the edge $3 \to \ell$, yielding the four-node chain $1 \to 2 \to 3 \to \ell$. Then $N = 4$, $N - 1 = 3$, and descendant counts are $3, 2, 1, 0$, so $C_R^{\max} = 1$ and

$$\mathrm{GRC}(h^*) = \frac{1}{3}\left[(1-1) + \left(1-\tfrac{2}{3}\right) + \left(1-\tfrac{1}{3}\right) + (1-0)\right] = \frac{2}{3} < \frac{3}{4} = \mathrm{GRC}(h),$$

so an admissible downward extension decreases the index, violating DE.

Concrete violation of UE*. Let $h_o = h$ be the three-node chain $1 \to 2 \to 3$, so $\mathrm{GRC}(h_o) = \mathrm{GRC}(h) = \frac{3}{4}$ and the weak dominance premise $\mathrm{GRC}(h_o) \geq \mathrm{GRC}(h)$ holds

with equality. Form $h^* \in h_o \uplus h$ by adding a single leaf-to-root cross-edge from the leaf of the upper copy to the root of the lower copy, yielding a six-node chain. For an N-node chain, GRC $= \frac{N}{2(N-1)}$ (since $C_R^{\max} = 1$ and $\sum_v (1 - C_R(v)) = \sum_{k=0}^{N-1} \frac{k}{N-1} = \frac{N}{2}$), so $\mathrm{GRC}(h^*) = \frac{6}{10} = \frac{3}{5} < \frac{3}{4} = \mathrm{GRC}(h)$. Thus UE* fails. (The same construction, with roles reversed, yields failure of DE*.)

Additivity, normalization, and separability. Because $C_R^{\max}$ depends on the entire hierarchy, GRC cannot admit a separable representation of the form required by AS, ADD, or QS. Finally, since the formula requires $N \geq 2$, GRC is undefined for a single isolated node, so INN cannot be evaluated.

Proposition 5. *The global reaching centrality index* GRC *satisfies* A *and* CRP*, and violates* IRP*,* ERP*,* SR*,* AS*,* ADD*,* QS*,* UE*,* DE*,* UE*, and* DE*. Moreover,* GRC *is undefined for a single isolated node, so* INN *cannot be evaluated.*

5.3 Krackhardt's graph theoretical dimensions

Krackhardt's (1994) four dimensions are best understood as diagnostics of connectedness, cyclicity, and redundancy rather than as depth indices. In our domain of directed acyclic graphs, this makes some of them either trivial or structurally incompatible with the axioms.

5.3.1 Connectedness

Formally, within our framework, the *connectedness index* for a hierarchy $h \in \mathscr{H}$ is

$$C(h) = \frac{|\{(i,j) \in V(h) \times V(h) : i \neq j \text{ and there exists an undirected path between } i \text{ and } j\}|}{n(n-1)},$$

where $n = |V(h)|$. Let $S(h)$ denote the number of ordered pairs (i, j) in the same weakly connected component. For non-trivial connected hierarchies ($n \geq 2$), $C(h) = 1$, while for disconnected hierarchies with $n \geq 2$, $0 \leq C(h) < 1$, with $C(h) = 0$ if and only if all weak components are singletons.

Location in our framework. Because connectedness relies on a global property of graph connectivity that cannot be decomposed into bilateral environment evaluations, C falls outside the separable families characterized in our framework. It does, however, satisfy anonymity (A).

Replication principles. Replication strictly alters connectivity by creating disjoint components. For a k-fold replication h^k, undirected paths only exist between nodes within the same copy. Thus $S(h^k) = kS(h)$, while the population size becomes kn. This yields the scaling identity

$$C(h^k) = \frac{kS(h)}{kn(kn-1)} = \frac{n-1}{kn-1}\frac{S(h)}{n(n-1)} = \frac{n-1}{kn-1}C(h).$$

Since $E(h) \neq \emptyset$ implies $n \geq 2$ and the existence of a weakly connected component of size at least 2, we have $S(h) > 0$ and hence $C(h) > 0$. Because $\frac{n-1}{kn-1} < 1$ for $k > 1$ and $n \geq 2$, the index strictly decreases under replication, satisfying CRP. Conversely, this identity demonstrates that C violates IRP (which requires invariance) and ERP (which requires an increase).

Monotonicity and extension axioms. C violates SR because it is insensitive to the removal of non-redundant edges that do not disconnect the underlying undirected graph. For example, consider a three-node chain $1 \to 2 \to 3$ where $C(h) = 1$. Removing the non-redundant edge $(2, 3)$ and adding $(1, 3)$ (the SR operation) results in a structure with edges $1 \to 2$ and $1 \to 3$. Because the graph remains weakly connected, C remains 1, violating the required strict decrease.

Similarly, if h is weakly connected (so $C(h) = 1$) and h_o is any weakly connected hierarchy with $|V(h_o)| \geq 2$ (so $C(h_o) = 1$), then every admissible compound h^* under UE or DE is weakly connected and therefore satisfies $C(h^*) = 1 = C(h)$, contradicting the required strict increase. The same example violates UE* and DE* since their weak-dominance premise $C(h_o) \geq C(h)$ holds with equality.

Additivity, normalization, and separability. C violates AS, ADD, and QS. This is confirmed by the replication scaling derived above: AS would require $C(h^k) = C(h)$, while QS would require $C(h^k) = \frac{1}{k}C(h)$. Similarly, ADD would imply linear scaling under replication (and hence ERP), which contradicts $C(h^k) = \frac{n-1}{kn-1}C(h)$. Finally, since the denominator evaluates to 0 when $n = 1$, C is undefined for a single isolated node, and INN cannot be evaluated.

Proposition 6. *The connectedness index C satisfies* A *and* CRP*, and violates* IRP*,* ERP*,* SR*,* AS*,* ADD*,* QS*,* UE*,* DE*,* UE***, *and* DE***. *Moreover, C is undefined for a single isolated node, so* INN *cannot be evaluated.*

5.3.2 Graph hierarchy

Krackhardt's *graph hierarchy index* is designed to measure the extent to which authority flows unidirectionally, penalizing mutual reachability (cyclic influence) between node pairs. Formally, for $h \in \mathscr{H}$ define

$$H(h) = 1 - \frac{P(h)}{\max P(h)},$$

where:

- $P(h)$ is the number of *unordered* pairs $\{i, j\}$ such that there exist directed paths $i \rightsquigarrow j$ *and* $j \rightsquigarrow i$ (mutual reachability);
- $\max P(h)$ is the number of *unordered* pairs $\{i, j\}$ such that there exists *at least one* directed path between them (either $i \rightsquigarrow j$ or $j \rightsquigarrow i$).

Thus $H(h) = 1$ corresponds to "perfect hierarchy" (no mutually reachable pairs), while $H(h) = 0$ corresponds to the extreme case in which every reachable pair is mutually reachable.

Triviality on DAGs. On directed acyclic graphs, mutual reachability between distinct nodes is impossible: if $i \rightsquigarrow j$ and $j \rightsquigarrow i$, then there is a directed cycle. Hence for every $h \in \mathscr{H}$, we have $P(h) = 0$. Therefore, whenever $\max P(h) > 0$ (i.e., whenever h contains at least one directed path), the index collapses to the constant value $H(h) = 1$. Consequently, on our domain of hierarchies with at least one subordination relation, Krackhardt's graph hierarchy index is constant and hence vacuous as a hierarchy measure: it cannot distinguish between acyclic hierarchies that differ in depth, span, or overall reporting architecture.

Location in our framework. Because H is constant on the non-flat part of $\mathscr{H}$ (whenever it is defined), it falls outside the scope of our depth-sensitive characterizations: any axiom requiring a strict response to structural change must fail. On its domain of definition, H does satisfy the label-invariance axiom A (vacuously, as a constant).

Replication principles. For any $h \in \mathscr{H}$ with $\max P(h) > 0$, we have $H(h) = 1$ and likewise $H(h^k) = 1$ for all $k \geq 1$. Thus, H satisfies the intensive replication principle IRP on its domain. However, this compliance is degenerate and trivially satisfied: it is driven strictly by the index collapsing to a constant value on directed acyclic graphs rather than functioning as a genuine structural average. Furthermore, because it does not strictly increase or strictly decrease when $E(h) \neq \varnothing$, it violates ERP and CRP there.

Population-growth (PG) regularity axioms. The regularity conditions IPG, EPG, and CPG are designed to enforce nontrivial responses to macroscopic structural growth within the average, total, and concentration paradigms, respectively. Since H is constant on its domain, these conditions are not applicable (and in particular cannot be satisfied in any nonvacuous sense).

Monotonicity and extension axioms. Constancy implies immediate failure of all strict monotonicity requirements. In particular, H violates SR (it cannot strictly decrease after removing a non-redundant edge), and it violates the strict extension axioms UE and DE, as well as their weak-dominance variants UE* and DE*, because every admissible transformation within $\mathscr{H}$ preserves acyclicity and hence leaves H equal to 1 whenever the index is defined.

Additivity, normalization, and separability. While one can represent a constant index in a vacuous separable form, H is undefined whenever $\max P(h) = 0$. In particular, for a single isolated node (and, more generally, for any edgeless hierarchy) we have $\max P(h) = 0$ and the expression above is undefined. Hence isolated node normalization INN cannot be evaluated for H.

Proposition 7. *On the class of hierarchies with at least one directed path (equivalently,* $\max P(h) > 0$*), Krackhardt's graph hierarchy index satisfies* $H(h) = 1$ *for all* $h \in \mathscr{H}$*. Consequently, the induced order* $\succcurlyeq_H$ *is constant on this domain and provides no nontrivial ranking of acyclic hierarchies; in particular, it fails the strict depth-sensitive axioms of our framework (including* SR*,* UE*,* DE*,* UE*, and* DE*). Moreover,* H *is undefined whenever* $\max P(h) = 0$ *(including the single-node hierarchy), so* INN *cannot be evaluated.*

5.3.3 Graph efficiency

Krackhardt's *graph efficiency index* evaluates how redundant a hierarchy's reporting links are relative to the minimum needed for weak connectivity. As such, it is a redundancy diagnostic rather than a depth-based hierarchy measure.

Definition. Let $h \in \mathscr{H}$ have weakly connected components $c = 1, \ldots, k$, where component c has n_c nodes and m_c directed edges (so $\sum_c n_c = |V(h)|$ and $\sum_c m_c = |E(h)|$). Krackhardt's

efficiency index is

$$E(h) = 1 - \frac{\sum_{c=1}^{k} \big(m_c - (n_c - 1)\big)}{\sum_{c=1}^{k} \frac{(n_c-1)(n_c-2)}{2}}.$$

The numerator counts *actual redundancy*: links in excess of the $n_c - 1$ undirected edges needed to keep component c weakly connected. The denominator counts *maximum possible redundancy* in component c relative to a complete undirected graph. We evaluate $E(h)$ only when the normalization is well-defined, i.e. when every component satisfies $n_c \geq 3$; otherwise the index is undefined.

Location in our framework. The index depends only on component-level counts (n_c, m_c) and ignores how authority is routed (depth, span, and reachability patterns). It therefore does not belong to the bilateral (depth–span) families characterized by our separability axioms. It does, however, satisfy anonymity A since relabeling preserves each (n_c, m_c) pair.

Replication principles. Let h^k denote the k-fold replication (disjoint union of k copies) of h. Replication multiplies both the numerator and denominator in the definition of E by k, so $E(h^k) = E(h)$ whenever both sides are defined. Hence E satisfies the intensive replication principle IRP (and thus RP). Since it does not strictly increase (resp. strictly decrease) under replication on non-flat hierarchies, it violates ERP and CRP on their intended domains.

Population-growth (PG) regularity axioms. The growth conditions IPG, EPG, and CPG are evaluated within the average, total, and concentration paradigms, respectively, and are therefore not applicable to E.

Monotonicity and extension axioms. The index violates SR because subordination removal can change reporting structure without changing component sizes or edge counts. For example, let h be the three-node chain $j \to i \to k$. Removing the non-redundant edge (i, k) and applying the SR shortcutting rule yields h' with edges $j \to i$ and $j \to k$. Both graphs have $n = 3$ and $m = 2$, hence $E(h) = E(h') = 1$, contradicting the strict part of SR.

The same structural blindness implies failure of the extension axioms. For instance, let h be the three-node chain $j \to i \to k$ (so $E(h) = 1$), let h_o be a single isolated node ℓ, and form an admissible upward extension $h^* \in h_o \oplus h$ by attaching ℓ above the unique root of h, yielding the four-node chain $\ell \to j \to i \to k$. Then h^* has $n = 4$ and $m = 3$, so it

still has zero redundancy and $E(h^*) = 1 = E(h)$, violating UE. An analogous construction (attaching an isolated node below the maximum-depth leaf) yields a violation of DE.

For the relaxed axioms, take $h_o = h$ equal to the same three-node chain (so $E(h_o) = E(h) = 1$), and form a partial compound $h^* \in h_o \uplus h$ by adding a single cross-edge from a leaf of the upper copy to a root of the lower copy. The resulting six-node graph is weakly connected with exactly $5 = n - 1$ edges, hence has zero redundancy and $E(h^*) = 1 = E(h)$, contradicting the strict conclusion of UE* (and symmetrically of DE*).

Additivity, normalization, and separability. Because E is constructed from component-level edge counts rather than node environments, it is incompatible with the node-level separability paradigms AS and ADD in our framework. Moreover, it also violates quadratic separability QS: any QS index dilutes under replication ($I(h^k) = \frac{1}{k} I(h)$), whereas $E(h^k) = E(h)$ whenever both sides are defined. Finally, E is undefined whenever some component has $n_c \leq 2$, which prevents evaluation of isolated node normalization INN in general.

Proposition 8. *The graph efficiency order $\succcurlyeq_E$ satisfies* A *and* IRP *(and hence* RP*), and violates* ERP*,* CRP*,* SR*,* AS*,* ADD*,* QS*,* UE*,* DE*,* UE**, *and* DE**. *Moreover, E is undefined whenever a component has at most two nodes, so* INN *cannot be evaluated.*

5.3.4 Least upper boundedness

Krackhardt's *least upper boundedness* (LUB) index measures how closely each weakly connected component of a hierarchy approximates a tree by checking whether pairs of nodes admit a *unique* least common boss.

Let $h \in \mathscr{H}$ have weakly connected components $c = 1, \ldots, k$, where component c has n_c nodes. Define

$$\mathrm{LUB}(h) = \frac{|\{(i,j) \in V(h) \times V(h) : i \neq j,\ i \text{ and } j \text{ lie in the same component, and } \exists!\ \mathrm{LUB}(i,j)\}|}{\sum_{c=1}^{k} n_c(n_c - 1)},$$

whenever the denominator is nonzero (equivalently, whenever at least one component has size ≥ 2). Here, for nodes i, j in the same component, let

$$B(i,j) = \{b \in V(h) : b \rightsquigarrow i \text{ and } b \rightsquigarrow j\},$$

where $\rightsquigarrow$ denotes reachability by a directed path and, by convention, each node reaches itself. Let $M(i,j)$ be the set of *minimal* elements of $B(i,j)$ under reachability (i.e., $b \in B(i,j)$ is

minimal if there is no distinct $b' \in B(i, j)$ with $b \rightsquigarrow b'$). A unique least upper bound exists iff $M(i, j)$ is a singleton.

Location in our framework. Because LUB is defined by global pair counts and the existence/uniqueness of common-boss sets, it does not belong to the bilateral (depth–span) families characterized by our separability axioms. It does, however, satisfy anonymity A since relabeling preserves reachability and least-upper-bound structure.

Replication principles. Let h^t denote the t-fold replication (disjoint union of t copies) of h. Replication multiplies both the numerator and denominator in the definition of LUB by t, hence $\mathrm{LUB}(h^t) = \mathrm{LUB}(h)$ whenever both sides are defined. Therefore LUB satisfies the intensive replication principle IRP, and it violates the strict replication variants ERP and CRP on their intended domains.

Population-growth (PG) regularity axioms. The growth conditions IPG, EPG, and CPG are evaluated within the average, total, and concentration paradigms, respectively, and are therefore not applicable to LUB.

Monotonicity and extension axioms. The index violates SR. Let h be the three-node chain $j \to i \to k$. Removing the non-redundant edge (i, k) and applying the SR shortcutting rule yields h' with edges $j \to i$ and $j \to k$. In both graphs, every ordered pair of distinct nodes has a unique least upper bound (indeed, $\mathrm{LUB}(h) = \mathrm{LUB}(h') = 1$), contradicting the strict part of SR.

The same insensitivity implies failure of the extension axioms. For instance, let h be the three-node chain $j \to i \to k$ (so $\mathrm{LUB}(h) = 1$), and let h_o be the two-node chain $\ell \to r$ (so $\mathrm{LUB}(h_o) = 1$). Any admissible compound $h^* \in h_o \oplus h$ that attaches the leaf r to the root j yields a longer chain, hence still satisfies $\mathrm{LUB}(h^*) = 1 = \mathrm{LUB}(h)$, violating UE. A symmetric construction attaching a two-node chain below the maximum-depth leaf yields a violation of DE.

For the relaxed axioms, take $h_o = h$ (so $\mathrm{LUB}(h_o) \geq \mathrm{LUB}(h)$ holds with equality) and form a partial compound $h^* \in h_o \uplus h$ by adding a single leaf-to-root cross-edge. The resulting hierarchy remains an arborescence, so $\mathrm{LUB}(h^*) = 1 = \mathrm{LUB}(h)$, contradicting the strict conclusions of UE* and DE*.

Additivity, normalization, and separability. Because LUB is defined by global pair counts rather than node environments, it violates AS and ADD. It also violates quadratic separability QS: any QS index dilutes under replication ($I(h^t) = \frac{1}{t} I(h)$), whereas $\mathrm{LUB}(h^t) = \mathrm{LUB}(h)$ whenever defined. Finally, LUB is undefined whenever $\sum_{c=1}^{k} n_c(n_c - 1) = 0$ (in particular, for a single isolated node), so isolated node normalization INN cannot be evaluated.

Proposition 9. *The least upper boundedness index* LUB *satisfies* A *and* IRP*, and violates* ERP*,* CRP*,* SR*,* AS*,* ADD*,* QS*,* UE*,* DE*,* UE***, *and* DE***. *Moreover,* LUB *is undefined whenever* $\sum_{c=1}^{k} n_c(n_c - 1) = 0$ *(including the single-node hierarchy), so* INN *cannot be evaluated.*

6 Concluding remarks

This paper transitions from partial hierarchical pre-orders to a complete axiomatic characterization of three numerical index families. Grounding measurement in structural principles, we demonstrate that all three families share a common foundation of additive decomposition. They diverge solely in how they normalize by organizational size, and this normalization choice is inseparably linked to the replication principle that governs each paradigm: absolute scaling defines an extensive view, linear normalization an intensive one, and quadratic normalization a concentration paradigm. No single index can satisfy more than one replication principle, making the choice of measurement lens fundamentally normative rather than technical.

This trichotomy resolves the tensions revealed by our impossibility results. We prove that normalized separability axioms cannot coexist with strict vertical extension: population averaging dilutes the index when flat structures are appended, while quadratic normalization collapses density unconditionally. Even weakened partial extension axioms fail under quadratic scaling, as the mathematical void created by structural fragmentation cannot be offset by finite local linkages. Resolving these tensions requires accepting the trade-off embedded in the replication principles. Under intensive normalization, we characterize the average bilateral family, where hierarchy scales as a per-capita average invariant under replication. These indices satisfy relaxed extension and subordination removal but necessarily sacrifice strict monotonicity to avoid dilution. Under extensive normalization, the total bilateral family sums node-level evaluations without a population denominator. Immune to dilution, these indices satisfy strict extension and subordination removal while scaling linearly with replication. Under quadratic normalization, the concentration family evaluates

authority density relative to potential dyadic structure. Because standard organizational expansion is inherently sparse, any genuine density metric must penalize vertical extensions; consequently, this family satisfies subordination removal and strict concentration under replication but necessarily violates all extension axioms.

Because each paradigm corresponds to a distinct replication principle, the characterization theorems provide uniquely identified tools tailored to the analyst's normative stance. Applying this lens to established indices reveals systematic axiom violations: most prominent measures either ignore directionality, collapse to constants on acyclic graphs, or rely on global normalization that violates structural separability. Among those examined, only the average depth index aligns with the core properties of a well-founded hierarchy measure. Future research can leverage these characterizations empirically, testing how different organizational contexts align with intensive versus extensive scaling, while further exploring the empirical implications of alternative functional specifications for local depth and span evaluations.

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